

Probabilistic Risk Analysis and Bayesian Decision Theory

Marcel van Oijen*

Mark Brewer†

```
knitr::opts_chunk$set( collapse=TRUE, comment=">" )  
library(copula)  
library(DiagrammeR)  
library(DiagrammeRsvg)  
library(fields)  
library(MCMCpack)  
library(mvtnorm)  
library(nimble)  
library(plot.matrix)  
library(readxl)  
library(rsvg)  
library(scales)  
library(terra)  
library(truncnorm)  
library(VineCopula)
```

*Independent researcher, Edinburgh, UK - VanOijenMarcel@gmail.com

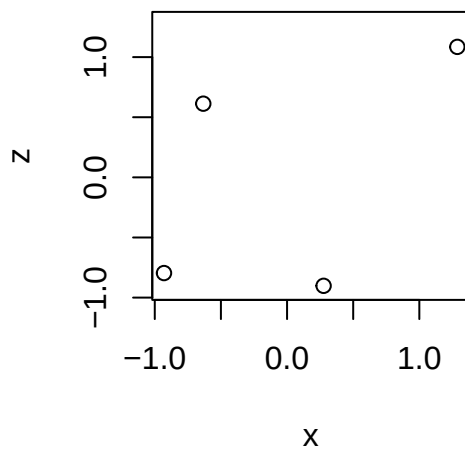
†BioSS Office, The James Hutton Institute, Craigiebuckler, Aberdeen AB15 8QH

Data

Sparse Dataset: {x₄,z₄}

```
set.seed(1)

n_4 <- 4 ; Sigma_4 <- matrix( c(1,0.5,0.5,1), nrow=2 )
xz_4 <- rmvnorm( n_4, c(0,0), Sigma_4 ) %*% chol( Sigma_4 )
xz_4 <- sweep( xz_4, 2, colMeans(xz_4) )
xz_4 <- xz_4 %*% solve( chol(cov(xz_4)) ) %*% chol(Sigma_4)
x_4 <- xz_4[,1] ; z_4 <- xz_4[,2]
m_4 <- colMeans(xz_4) ; S_4 <- cov(xz_4)
```



A Collection of Linear Datasets: l_{xz}.L

```
set.seed(1)

mu <- c(0,0) ; Sigma <- diag(1,2) ; Sigma[1,2] <- Sigma[2,1] <- 0.5
n_d <- 1e2 ; l_xz.L <- vector("list",n_d) ; n <- 1e3
for(d in 1:n_d) { l_xz.L[[d]] <- rmvnorm( n, mu, Sigma ) }
```

A Collection of Nonlinear Datasets: l_{xz}.NL

```
set.seed(1)

n_d <- 1e2 ; l_xz.NL <- vector("list",n_d) ; n <- 1e3 ; sz <- 0.1
for(d in 1:n_d) {
  x <- runif( n, 0, 3 ) ; ez <- rnorm( n, 0, sz ) ; z <- 1-exp(-x) + ez
  l_xz.NL[[d]] <- cbind(x,z) }
```

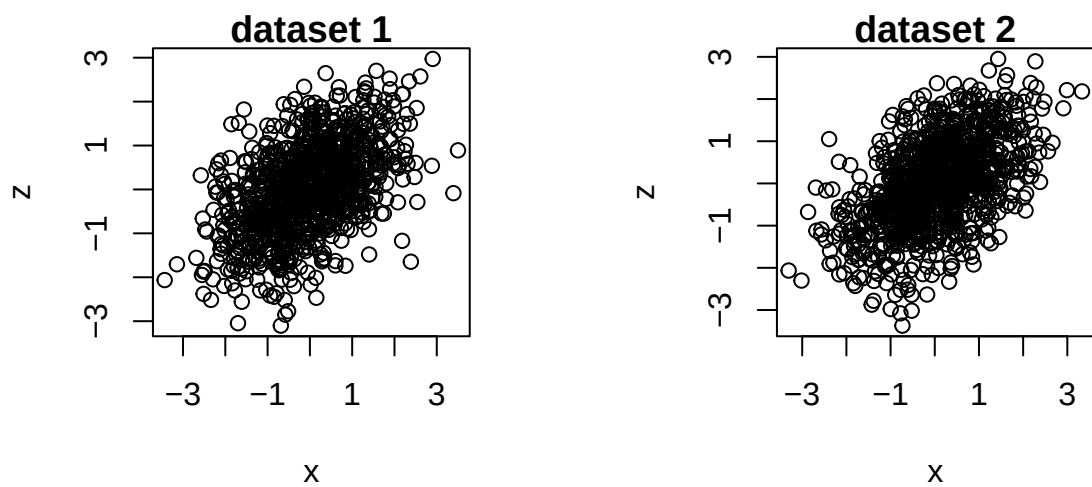


Figure 1: The first two data sets generated from a noisy linear relationship.

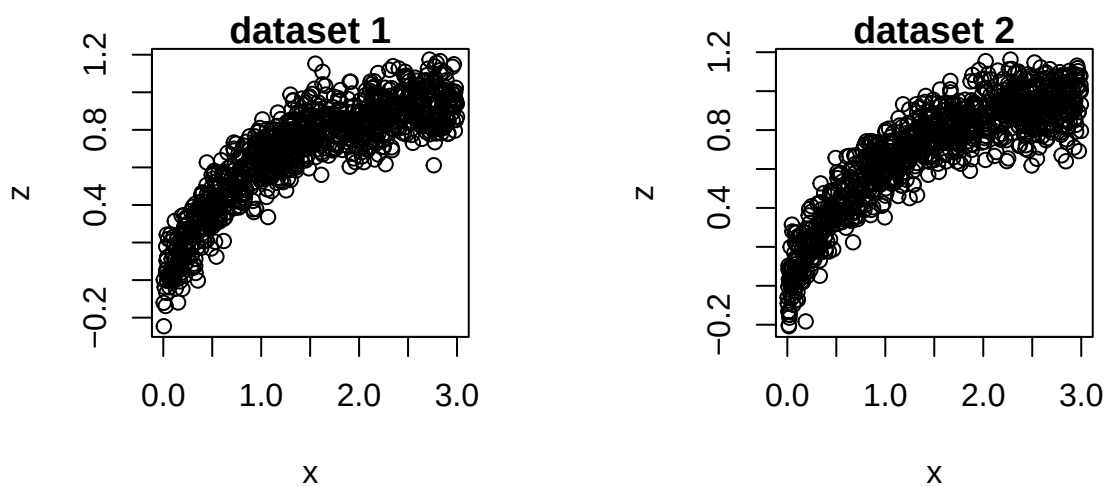


Figure 2: The first two data sets generated from a noisy negatively exponential relationship.

German Forestry Data: x_r3, z_Fs, z_Q, z_Pa, z_Ps

```

d.data      <- "data_Germany/"

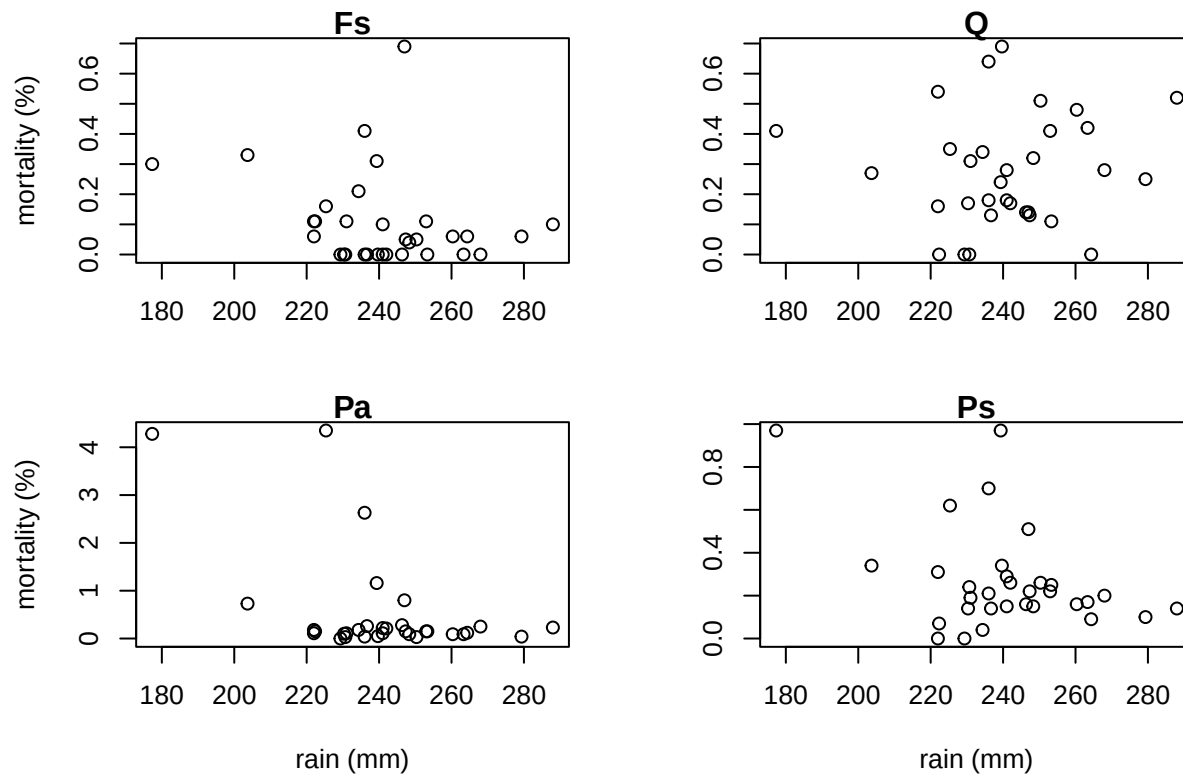
file.data   <- paste0( d.data, "2-bis-7_abb-tab_nsh_2024-04-08.xlsx" )
sheet.data  <- "5_Daten"
r.data      <- "B120:C153"
y_r         <- read_excel( file.data, sheet=sheet.data, range=r.data, col_names=F )
> New names:
> * `` -> `...1`
> * `` -> `...2`
y_r         <- round( as.matrix(y_r) )
colnames(y_r) <- c( "year", "rain.mm" )

file.data   <- paste0( d.data, "absterberate_EI_GFI_GKI_RBU_zeitreihe.csv" )
y_m         <- read.csv2( file.data )
colnames(y_m) <- c( "year", "mort.-%_Fs", "mort.-%_Q", "mort.-%_Pa", "mort.-%_Ps" )
y_r_m       <- cbind( y_r, y_m[ , startsWith( colnames(y_m), "mort" ) ] )

# Annual mortality vs. summer rain of last three years
meanlast <- function(v,k){as.vector(filter(v,f=rep(1/k,k),s=1))}
n        <- dim(y_r_m)[1] ; k <- 3
x_r3     <- meanlast(y_r_m[, "rain.mm"], k)[k:n]

par ( mfrow=c(2,2), mar=c(4,4,1,2) )
z_Fs <- y_r_m[k:n, "mort.-%_Fs"]
plot( x_r3, z_Fs, main="Fs", xlab="", ylab="mortality (%)" )
z_Q  <- y_r_m[k:n, "mort.-%_Q"]
plot( x_r3, z_Q, main="Q", xlab="", ylab="" )
z_Pa <- y_r_m[k:n, "mort.-%_Pa"]
plot( x_r3, z_Pa, main="Pa", xlab="rain (mm)", ylab="mortality (%)" )
z_Ps <- y_r_m[k:n, "mort.-%_Ps"]
plot( x_r3, z_Ps, main="Ps", xlab="rain (mm)", ylab="" )

```



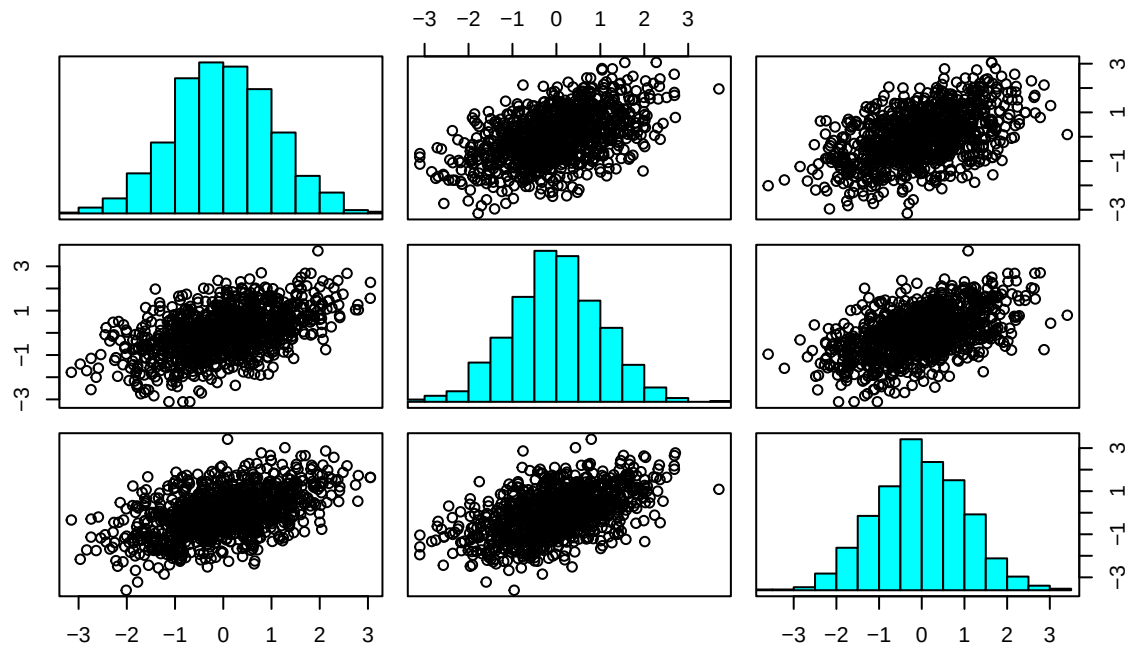
Trivariate Gaussian Dataset

```
panel.hist <- function(x, ...) {
  h      <- hist(x, plot=F)
  breaks <- h$breaks; n_b <- length(breaks)
  y      <- h$counts; y  <- min(x) + (max(x)-min(x)) * y/max(y)
  rect(breaks[-n_b], min(x), breaks[-1], y, col="cyan", ...) }

set.seed(1)

n_G3 <- 1e3 ; Sigma_G3 <- matrix(0.5,nrow=3,ncol=3) ; diag(Sigma_G3) <- 1
xz_G3 <- rmvnorm( n_G3, c(0,0,0), Sigma_G3 )

pairs(xz_G3, diag.panel=panel.hist, label="" ) ; cor(xz_G3)
```



```

>           [,1]      [,2]      [,3]
> [1,] 1.0000000 0.4993916 0.4933134
> [2,] 0.4993916 1.0000000 0.5190580
> [3,] 0.4933134 0.5190580 1.0000000

```

Introduction to Probabilistic Risk Analysis (PRA)

From risk matrices to PRA

Table 1: Typical risk matrix.

Consequence Probability	1 (unlikely)	2 (medium)	3 (likely)
1 (minor)	1	2	3
2 (medium)	2	4	6
3 (major)	3	6	9

Basic equations for PRA

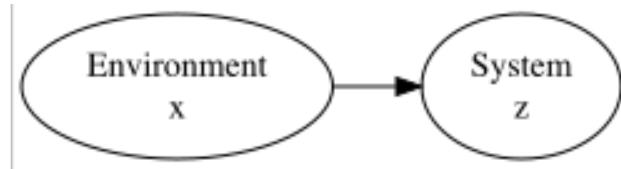


Figure 3: PRA as a probabilistic network.

$$R = E[z|\neg H] - E[z], \quad (1)$$

where $\neg H$ stands for non-hazardous environmental conditions.

$$\begin{aligned}
 V &= R / p[H] \\
 &= (E[z|\neg H] - E[z]) / p[H] \\
 &= (E[z|\neg H] - p[H]E[z|H] - (1 - p[H])E[z|\neg H]) / p[H] \\
 &= E[z|\neg H] - E[z|H].
 \end{aligned} \quad (2)$$

Decomposition of risk: 2 or 3 components

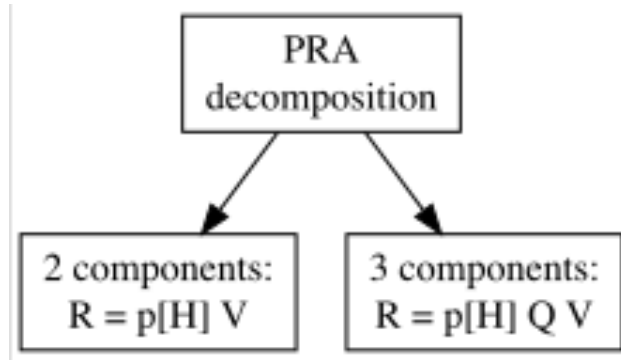


Figure 4: PRA-classification according to treatment of the hazardous region.

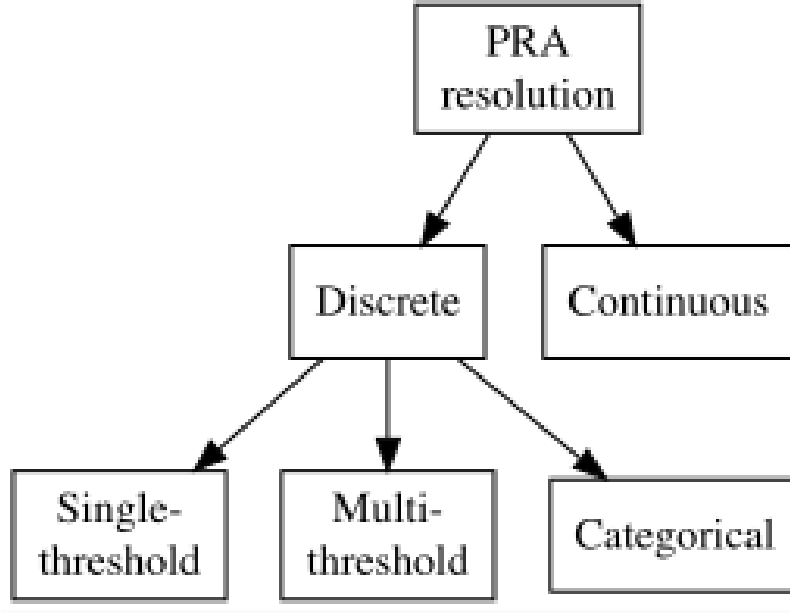


Figure 5: PRA-classification according to treatment of the hazardous region.

Resolution of PRA: single-threshold, multi-threshold, categorical, continuous

Single-threshold PRA

$$\begin{aligned}
 V &= E[z|x \geq thr] - E[z|x < thr], \\
 R &= E[z|x \geq thr] - E[z] \\
 &= p[H] V.(\#eq : PRAttr1)
 \end{aligned} \tag{3}$$

Multi-threshold PRA

$$\begin{aligned}
 p[H_i] &= p[thr_{i-1} \leq x < thr_i], \\
 V_i &= E[z|\neg H] - E[z|H_i] \\
 &= E[z|x \geq thr_n] - E[z|thr_{i-1} \leq x < thr_i], \\
 R_i &= p[H_i] V_i, (\#eq : PRAttrnB)
 \end{aligned} \tag{4}$$

for $i = 1, \dots, n$ and where $thr_0 = -\infty$.

$$\begin{aligned}
 p[H] &= \sum p[H_i], \\
 V &= \sum \frac{p[H_i]}{p[H]} V_i, \\
 R &= \sum R_i, \\
 &= p[H] V, (\#eq : PRAttrnA)
 \end{aligned} \tag{5}$$

with all summations running from $i = 1$ to n .

Categorical PRA

Continuous PRA

$$R = \int_{x=-\infty}^{thr} r(x)dx, (\#eq : PRAcontA) \quad (6)$$

where

$$\begin{aligned} r(x) &= p[x] v(x), \\ v(x) &= E[z|x \geq thr] - E[z|x].(\#eq : PRAcontB) \end{aligned} \quad (7)$$

Implementation of PRA: distribution-based, sampling-based, model-based

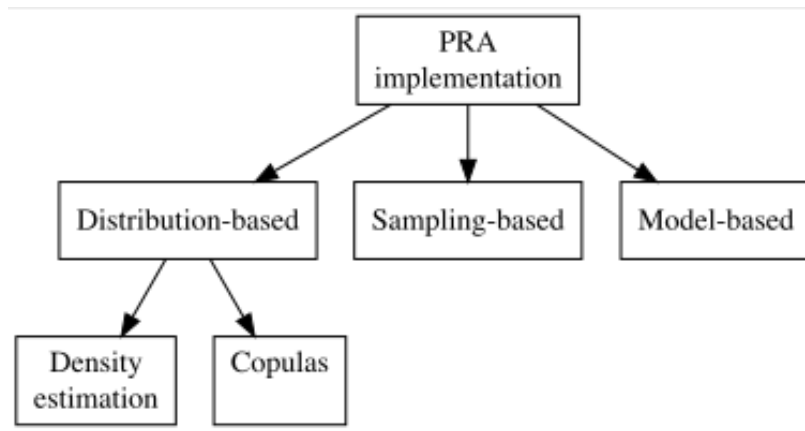


Figure 6: PRA-classification according to implementation method.

Distribution-based single-threshold PRA

Conditional distributions for z

$$\begin{aligned}
p[z|x < thr] &= \frac{p[z, x < thr]}{p[x < thr]} \\
&= \frac{p[z] \int_{x=-\infty}^{thr} p[x|z] dx}{p[x < thr]} \\
&= \frac{p[z] F_{x|z}[thr]}{F_x[thr]}, \\
p[z|x \geq thr] &= \frac{p[z](1 - F_{x|z}[thr])}{1 - F_x[thr]}, (\#eq : pzGxbelowabove)
\end{aligned} \tag{8}$$

where $F_x[thr]$ and $F_{x|z}[thr]$ are the cumulative distribution functions associated with $p[x]$ and $p[x|z]$, both evaluated at $x = thr$. Note that $F_x[thr] = p[x < thr]$ is the hazard probability $p[H]$.

$$\begin{aligned}
E[z|x < thr] &= \int_{z=-\infty}^{\infty} z p[z|x < thr] dz \\
&= \frac{1}{F_x[thr]} \int_{z=-\infty}^{\infty} z p[z] F_{x|z}[thr] dz, \\
E[z|x \geq thr] &= \frac{1}{1 - F_x[thr]} \int_{z=-\infty}^{\infty} z p[z] (1 - F_{x|z}[thr]) dz. (\#eq : EzGxbelowaboveA)
\end{aligned} \tag{9}$$

$$\begin{aligned}
E[z|x < thr] &= \frac{1}{F_x[thr]} \int_{x=-\infty}^{thr} p[x] E[z|x] dx, \\
E[z|x \geq thr] &= \frac{1}{1 - F_x[thr]} \int_{x=thr}^{\infty} p[x] E[z|x] dx. (\#eq : EzGxbelowaboveB)
\end{aligned} \tag{10}$$

Conditions for V being constant

$$\begin{aligned}
\frac{dV}{dthr} &= \frac{d(E[z|x \geq thr] - E[z|x < thr])}{dthr} \\
&= p[thr] \left\{ \frac{E[z|x \geq thr] - z[thr]}{1 - F_x[thr]} - \frac{z[thr] - E[z|x < thr]}{F_x[thr]} \right\} \\
&= \frac{p[thr]}{F_x[thr](1 - F_x[thr])} \{E[z|x \geq thr]F_x[thr] - z[thr] + E[z|x < thr](1 - F_x[thr])\}, (\#eq : EzdVdthr)
\end{aligned} \tag{11}$$

where $z[thr] = E[z|x = thr]$.

Example of distribution-based PRA: Gaussian $p[x, z]$

EXAMPLE: Bivariate Gaussian for x and z :

$$\begin{aligned}
p[x, z] &= N[\mu, \Sigma], \text{ where} \\
\mu &= \begin{bmatrix} \mu_x \\ \mu_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}; \quad \Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_z \\ \rho\sigma_x\sigma_z & \sigma_z^2 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}. (\#eq : pxzGaussian)
\end{aligned} \tag{12}$$

$$\begin{aligned}
p[x] &= N[\mu_x, \sigma_x^2] = N[0, 1], \\
p[z] &= N[\mu_z, \sigma_z^2] = N[0, 1], \\
p[z|x] &= N[\mu_z + \rho(x - \mu_x) \frac{\sigma_z}{\sigma_x}, \sigma_z^2(1 - \rho^2)] = N[0.5x, 0.75].
\end{aligned} \tag{13}$$

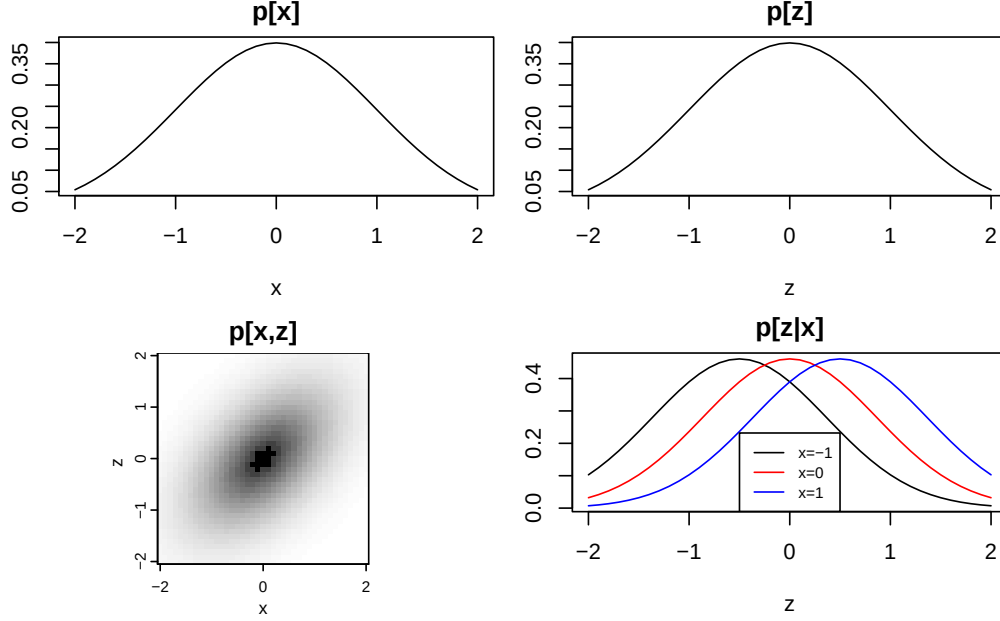


Figure 7: Marginal, joint, and conditional Gaussian distributions for x and z .

Hazard probability and conditional distributions

Conditional expectations and PRA

```

Ez_xlo_NI <- function(thr=0, mz.=mz, mx.=mx, Vz.=Vz, Vx.=Vx, r=rxz) {
  x.seq <- seq( thr-5*sqrt(Vx.), thr, length.out=101 )
  px.seq <- px( x.seq, m=mx., V=Vx. ) ; px.sum <- sum(px.seq)
  Ez_x.seq <- mz. + r * (x.seq-mx.) * sqrt(Vz./Vx.)
  Ezxbelow <- sum(px.seq * Ez_x.seq) / px.sum
  return( Ezxbelow ) }

```

Formulas for the conditional bivariate Gaussian expectations

EXPECTATIONS FOR BIVARIATE GAUSSIAN $p[x,z]$:

$$\begin{aligned}
E[z|x < thr] &= E[z] - \rho \sigma_x \sigma_z \frac{p_x[thr]}{F_x[thr]}, \\
E[z|x \geq thr] &= E[z] + \rho \sigma_x \sigma_z \frac{p_x[thr]}{1 - F_x[thr]}. (\#eq : EzcondGaussian)
\end{aligned} \tag{14}$$

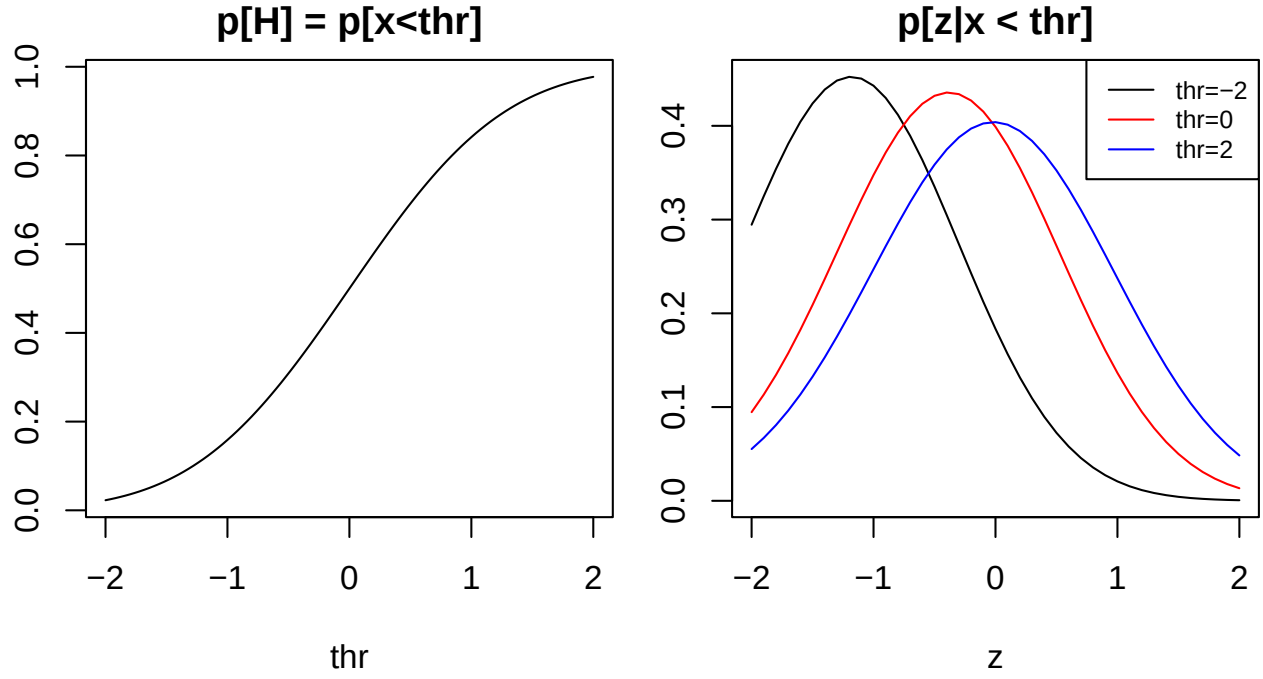


Figure 8: $p[H]$ and $p[z|x < \text{thr}]$ for the bivariate Gaussian.

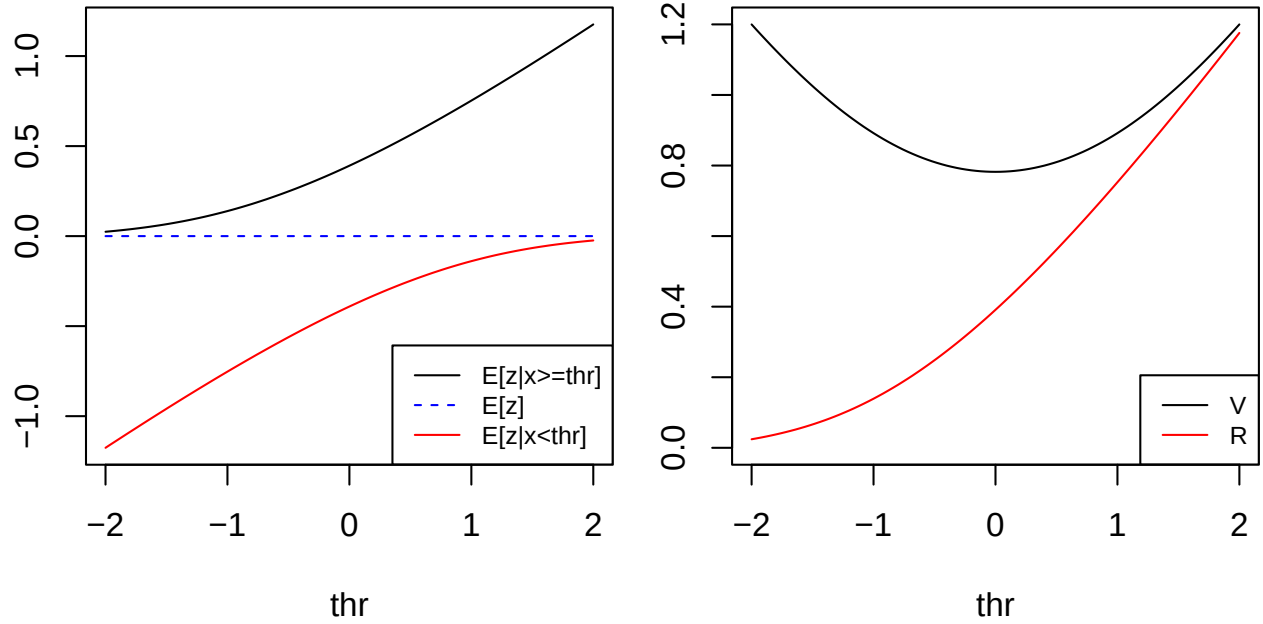


Figure 9: Distribution-based single-threshold PRA: bivariate Gaussian example. Left: conditional expectations for z as a function of threshold-value. Right: the corresponding values of V and R .

```

EzVz_Gauss <- function( m.=m, S.=S, thr.=thr ) {
  mx      <- m.[1]      ; mz <- m.[2]
  Vx      <- S.[1,1]    ; Vz <- S.[2,2] ; Vxz <- S.[1,2] ; r <- Vxz/sqrt(Vx*Vz)
  pthr    <- dnorm(thr., mx, sqrt(Vx)) ; Fthr <- pnorm(thr., mx, sqrt(Vx))
  Ez_xlo  <- mz - Vxz * pthr / Fthr
  Ez_xhi  <- mz + Vxz * pthr / (1-Fthr)
  Vz_xlo  <- Vz + r * (thr.-mx) * (Ez_xlo-mz) - (Ez_xlo-mz)^2
  Vz_xhi  <- Vz + r * (thr.-mx) * (Ez_xhi-mz) - (Ez_xhi-mz)^2
  result  <- c( Ez_xlo, Ez_xhi, Vz_xlo, Vz_xhi )
  names ( result ) <- c( "Ez_xlo", "Ez_xhi", "Vz_xlo", "Vz_xhi" )
  return( result ) }

```

```

xz <- l_xz.L[[1]] ; m <- colMeans(xz) ; mx <- m[1] ; mz <- m[2]
S <- var(xz) ; r <- S[1,2] / sqrt(Vx * Vz) ; Vx <- S[1,1] ; Vz <- S[2,2]

```

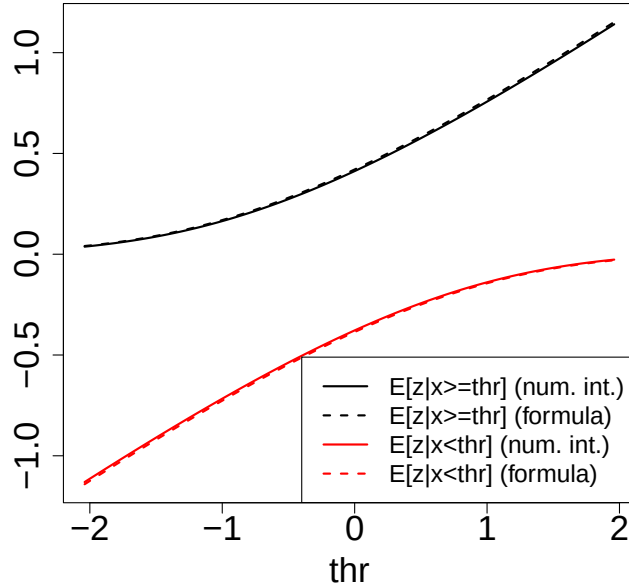


Figure 10: Conditional expectations for z as a function of threshold-value in the case of bivariate Gaussian $p[x, z]$. Solid lines: Numerical integration. Dashed lines: Formulas.

PRA FOR BIVARIATE GAUSSIAN $p[x, z]$:

$$\begin{aligned}
p[H] &= F_x[thr], \\
V &= \frac{\rho \sigma_x \sigma_z p_x[thr]}{F_x[thr] (1 - F_x[thr])}, \\
R &= \frac{\rho \sigma_x \sigma_z p_x[thr]}{1 - F_x[thr]}. (\#eq : PRAGaussian)
\end{aligned} \tag{15}$$

Sampling-based single-threshold PRA

SAMPLE MEANS:

$$\begin{aligned}\hat{E}[z] &= \bar{z}, \\ \hat{E}[z|x < thr] &= \bar{z}_H, \\ \hat{E}[z|x \geq thr] &= \bar{z}_{\neg H}.(\#eq : Esample)\end{aligned}\tag{16}$$

```
PRA0 <- function( x, z, thr=0 ) {
  n <- length(x) ; H <- which(x < thr) ; n_H <- length(H)
  Ez_H <- mean( z[ H] ) ; Ez_notH <- mean( z[ -H] )
  pH <- n_H / n ; V <- Ez_notH - Ez_H ; R <- pH * V
  return( c( pH=pH, V=V, R=R ) ) }
```

Example of sampling-based PRA: linear relationship

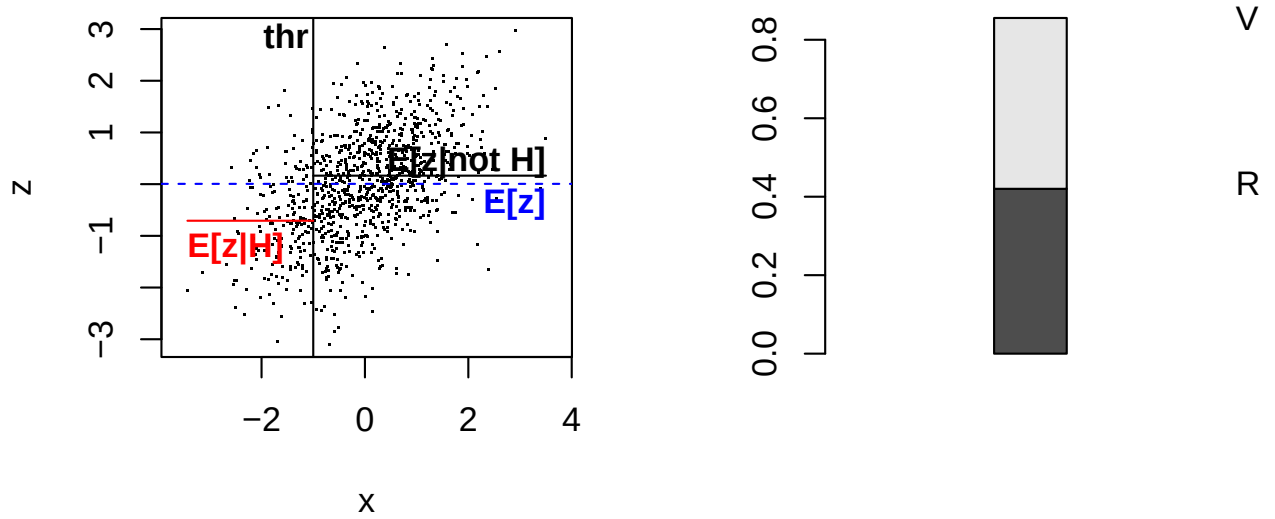


Figure 11: Sampling-based single-threshold PRA on 'data set 1' with $thr=0$. Left: data and expectation values. Right: V and R .

Varying the threshold

Example of sampling-based PRA: nonlinear relationship

NONLINEAR RELATIONSHIP BETWEEN x AND z :

$$\begin{aligned}z|x &\sim N[f(x), \sigma_z^2], \quad x \geq 0, \text{ where} \\ f(x) &= 1 - \exp(-x).(\#eq : negexp)\end{aligned}\tag{17}$$

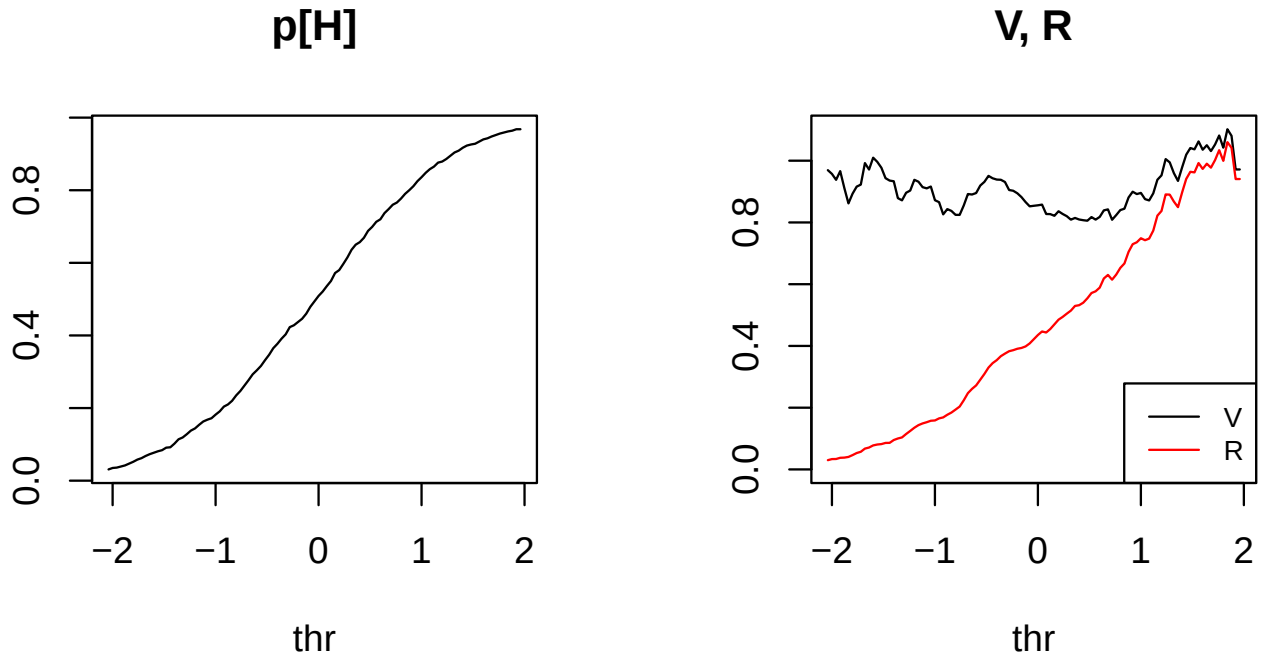


Figure 12: Sampling-based single-threshold PRA on data set 1 for a range of different thresholds. Left: $p[H]$. Right: V and R .

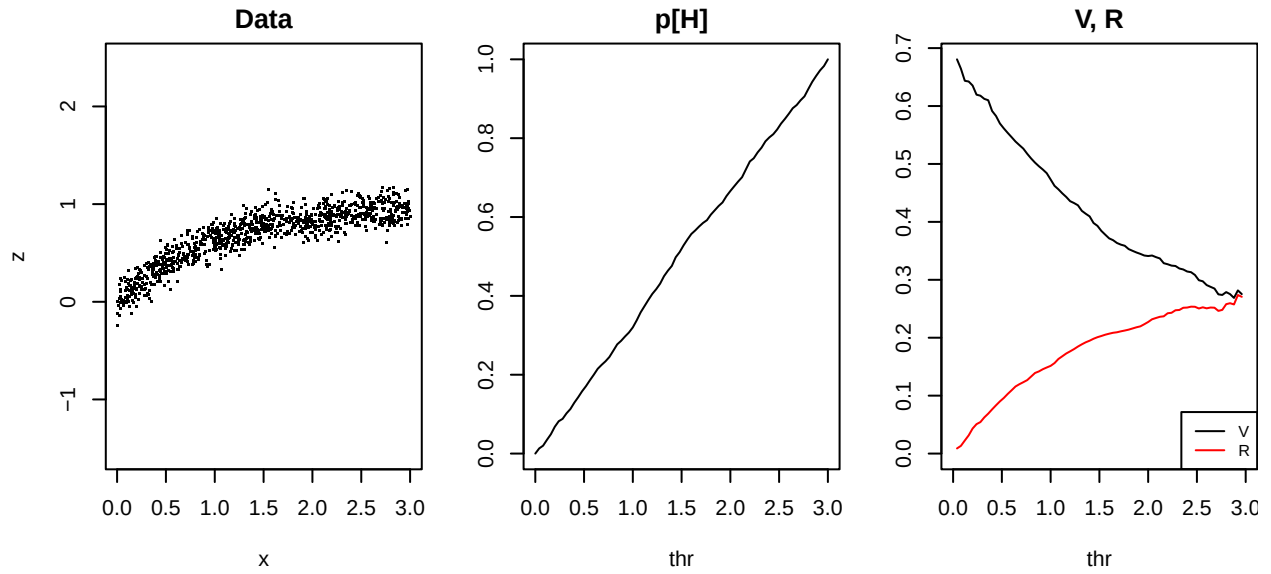


Figure 13: PRA on a data set from a nonlinear relationship.

Sampling-based single-threshold PRA: Uncertainty quantification (UQ)

STANDARD DEVIATIONS OF SAMPLE MEANS:

$$\begin{aligned}\sigma_{\hat{E}[z|x<thr]} &= \sqrt{\frac{Var[z_H]}{n_H}}, \\ \sigma_{\hat{E}[z|x\geq thr]} &= \sqrt{\frac{Var[z_{\neg H}]}{n - n_H}}. (\#eq : sigmasample)\end{aligned}\tag{18}$$

Uncertainty in $p[H]$

SAMPLING UNCERTAINTY FOR $p[H]$:

$$\sigma_{p[H]} = \sqrt{p[H] (1 - p[H]) / n}, (\#eq : sigmapH)\tag{19}$$

BAYESIAN POSTERIOR UNCERTAINTY FOR $p[H]$:

$$\begin{aligned}p[H] \sim Be[a, b] &\implies \\ \sigma_{p[H]} &= \frac{1}{a+b} \sqrt{\frac{ab}{a+b+1}}, (\#eq : sigmapHBayes)\end{aligned}\tag{20}$$

where $a = 1 + n_H$ and $b = 1 + n - n_H$.

Uncertainty in V

SAMPLING UNCERTAINTY FOR V :

$$\sigma_V = \sqrt{\sigma_{\hat{E}[z|x\geq thr]}^2 + \sigma_{\hat{E}[z|x<thr]}^2}, (\#eq : sigmaV)\tag{21}$$

where $\sigma_{\hat{E}[z|x\geq thr]}$ and $\sigma_{\hat{E}[z|x<thr]}$ are as defined in Eq. @ref(eq:sigmasample).

Uncertainty in R

SAMPLING UNCERTAINTY FOR R :

$$\sigma_R = \sqrt{\sigma_{p[H]}^2 \sigma_V^2 + \sigma_V^2 p[H]^2 + \sigma_{p[H]}^2 V^2}. (\#eq : sigmaR)\tag{22}$$

Extension of R-code for PRA: adding the UQ

```
PRA <- function( x, z, thr=0 ) {
  n      <- length(x)      ; H      <- which(x < thr) ; n_H <- length(H)
  Ez_H   <- mean( z[ H] ) ; s_Ez_H <- sqrt( var(z[ H]) / n_H )
  Ez_notH <- mean( z[-H] ) ; s_Ez_notH <- sqrt( var(z[-H]) / (n-n_H) )
  pH     <- n_H / n      ; V      <- Ez_notH - Ez_H ; R <- pH * V
  s_pH   <- sqrt( pH*(1-pH) / n )
  s_V    <- sqrt( s_Ez_H^2 + s_Ez_notH^2 )
  s_R    <- sqrt( s_pH^2*s_V^2 + s_pH^2*V^2 + pH^2*s_V^2 )
  return( c(pH=pH,V=V,R=R,s_pH=s_pH,s_V=s_V,s_R=s_R) )
}
```


PRA with UQ on the nonlinear data set

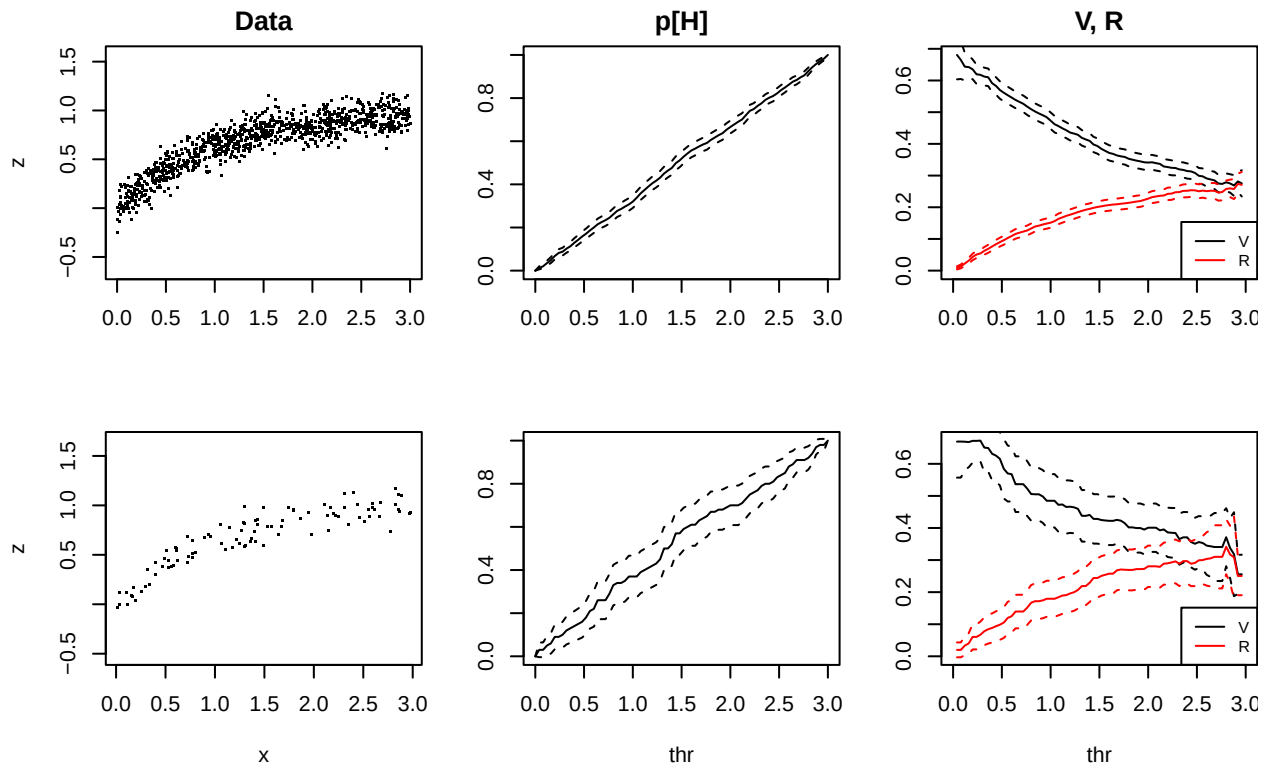


Figure 14: PRA on nonlinearly related x and z , with UQ. Top row: as previous example. Bottom row: same but just 10% of data being used. Dashed lines are 2 standard deviations away from the mean.

Verification of the UQ by simulating multiple data sets

UQ-verification: Nonlinear relationship

```
thr <- 1

PRA.tbl <- t( sapply( 1:length(l_xz.NL), function(d) {
  PRA( l_xz.NL[[d]][,1], l_xz.NL[[d]][,2], thr) } ) )

PRA.tbl[1:3,]
>      pH      V      R      s_pH      s_V      s_R
> [1,] 0.320 0.4732057 0.1514258 0.01475127 0.01298911 0.008126446
> [2,] 0.344 0.5008567 0.1722947 0.01502212 0.01328335 0.008805076
> [3,] 0.311 0.4681190 0.1455850 0.01463827 0.01250033 0.007880549

slist_pH <- sd( PRA.tbl[, "pH" ] )
slist_V  <- sd( PRA.tbl[, "V" ] )
slist_R  <- sd( PRA.tbl[, "R" ] )
```

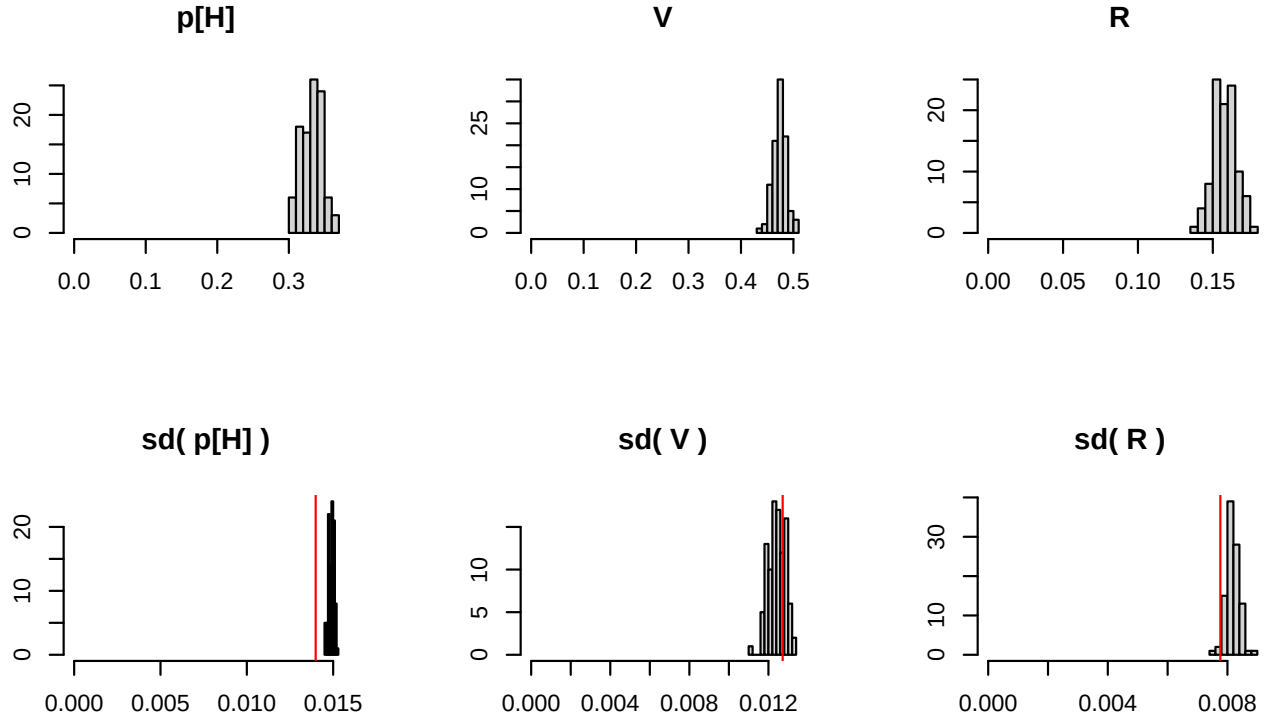


Figure 15: PRAs on 100 realisations (virtual data sets) from the same nonlinear relationship between x and z . Top row: distributions of the 100 estimates for $p[H]$, V and R . Bottom row: distributions of the 100 uncertainty estimates (sigma-values). The vertical red lines indicate the standard deviations of the top-row estimates (widths of top-row histograms).

UQ-verification: Linear relationship

```
>           pH          V          R          s_pH          s_V          s_R
> [1,] 0.182 0.8719316 0.1586916 0.01220148 0.07483537 0.01730677
> [2,] 0.143 1.0849957 0.1551544 0.01107028 0.07795630 0.01640994
> [3,] 0.166 0.9192960 0.1526031 0.01176622 0.07231166 0.01618065
```

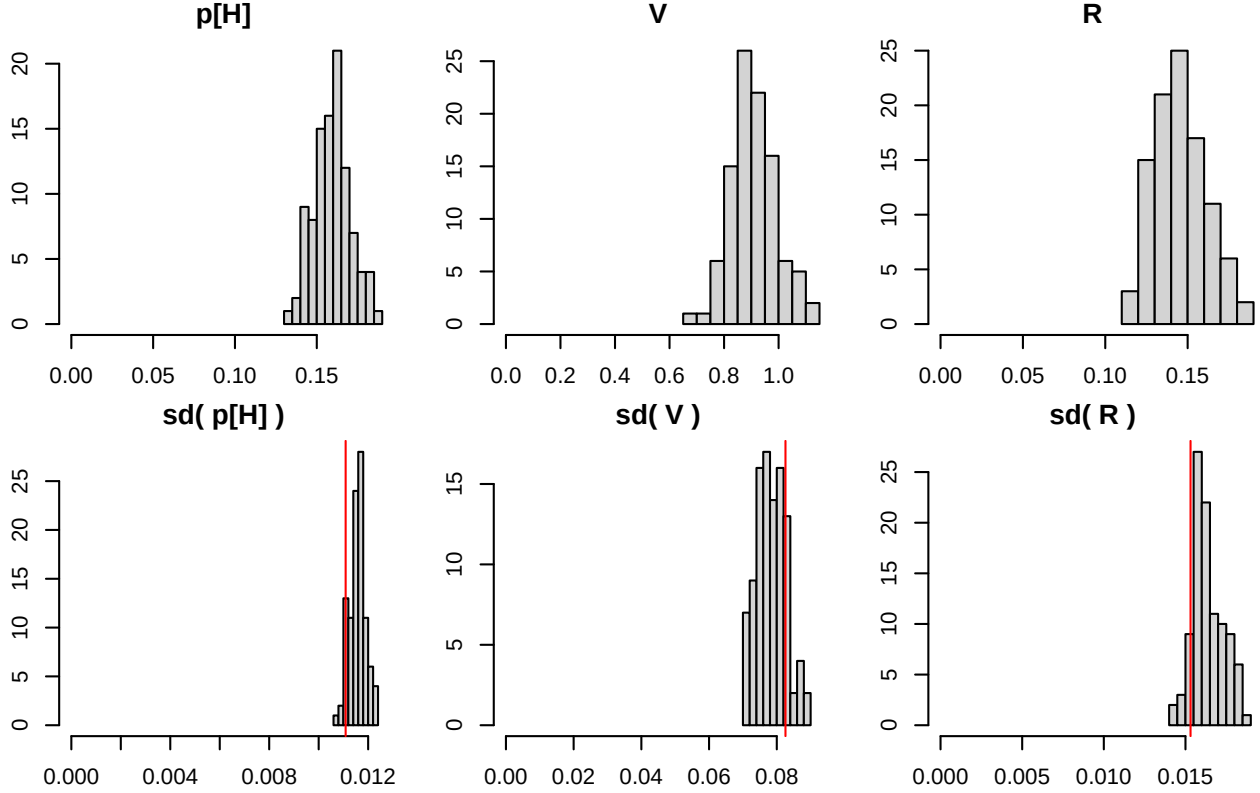


Figure 16: PRAs on 1000 realisations from the same linear relationship between x and z . Top row: 1000 estimates for $p[H]$, V and R . Bottom row: 1000 uncertainty estimates. Vertical red lines indicate standard deviations of the top-row estimates.

Formulas for the conditional bivariate Gaussian variances

VARIANCES OF BIVARIATE GAUSSIAN $p[x,z]$:

$$Var[z|x < thr] = \sigma_z^2 + \rho(thr - E[x])(\mu_1 - E[z]) - (\mu_1 - E[z])^2, \quad (23)$$

$$Var[z|x \geq thr] = \sigma_z^2 + \rho(thr - E[x])(\mu_2 - E[z]) - (\mu_2 - E[z])^2, (\#eq: VarzcondGaussian)$$

where $\mu_1 = E[z|x < thr]$ and $\mu_2 = E[z|x \geq thr]$ for which the formulas were given in Eq. @ref(eq:EzcondGaussian).

```
PRA0_Gauss <- function( m.=m, S.=S, thr.=thr ) {
  mx <- m.[1] ; sx <- sqrt(S.[1,1]) ; Vxz <- S.[1,2]
  pH <- pnorm( thr., mx, sx )
  V <- Vxz * dnorm(thr., mx, sx) / (pH * (1-pH))
  R <- pH * V
  return( c( pH=pH, V=V, R=R ) ) }
```

```

PRA_Gauss <- function( m.=m, S.=S, n.=n, thr.=thr ) {
  pH <- V <- R <- rep( NA, 1e3 )
  for(j in 1:1e3){
    S <- riwish( n.-1, S. * (n.-1) ) ; m <- rmvnorm( 1, m., S/n. )
    PRA <- PRAO_Gauss( m, S, thr. )
    pH[j] <- PRA["pH"] ; V[j] <- PRA["V"] ; R[j] <- PRA["R"]
  }
  return( c( pH=mean(pH), V=mean(V), R=mean(R),
             s_pH=sd (pH), s_V=sd (V), s_R=sd (R) ) )
}

```

```

>           pH           V           R           s_pH           s_V           s_R
> 0.1588730000 0.9077771702 0.1442212824 0.0003655576 0.0024716987 0.0005141248
>           pH           V           R           s_pH           s_V           s_R
> 0.1586450576 0.9063497560 0.1437879427 0.0002921984 0.0016671928 0.0003869262

```

Density estimation to move from sampling- to distribution-based PRA

```
xz <- l_xz.L[[1]] ; m <- colMeans(xz) ; V <- var(xz)
```

```
mx <- m[1] ; mz <- m[2]  
Vx <- V[1,1] ; Vz <- V[2,2] ; rxz <- V[1,2] / sqrt(Vx * Vz)  
thr <- -1  
Ez <- mz  
Ez_xlo <- Ez_xlo_NI(thr=thr, mz.=mz, mx.=mx, Vz.=Vz, Vx.=Vx, r=rxz)  
Ez_xhi <- Ez_xhi_NI(thr=thr, mz.=mz, mx.=mx, Vz.=Vz, Vx.=Vx, r=rxz)
```

```
V <- Ez_xhi - Ez_xlo  
R <- Ez_xhi - Ez  
pH <- R / V
```

```
> PRA using density estimation and eqs for conditional expectations:
```

```
> 0.1803815 0.8810575 0.1589265
```

```
>           pH           V           R           s_pH           s_V           s_R  
> 0.18200000 0.87193163 0.15869156 0.01220148 0.07483537 0.01730677
```

Copulas for distribution-based PRA

SKLAR's THEOREM:

$$F_{xz}(x, z) = C(F_x(x), F_z(z)), (\#eq : Sklar) \quad (24)$$

Sampling from copulas and carrying out PRA

```
cpN      <- normalCopula( param=0.5, dim=2 )
mvN.NN   <- mvdc( cpN, margins = c("norm", "norm"),
                  paramMargins = list( list( mean=0, sd=1 ),
                                       list( mean=2, sd=1 ) ) )
```

```
n        <- 1e3
sample.cpN <- rCopula( n, cpN )
Fx.N     <- sample.cpN[,1] ; Fz.N <- sample.cpN[,2]
sample.mvN.NN <- rMvdc ( n, mvN.NN )
x.N.NN    <- sample.mvN.NN[,1] ; z.N.NN <- sample.mvN.NN[,2]
```

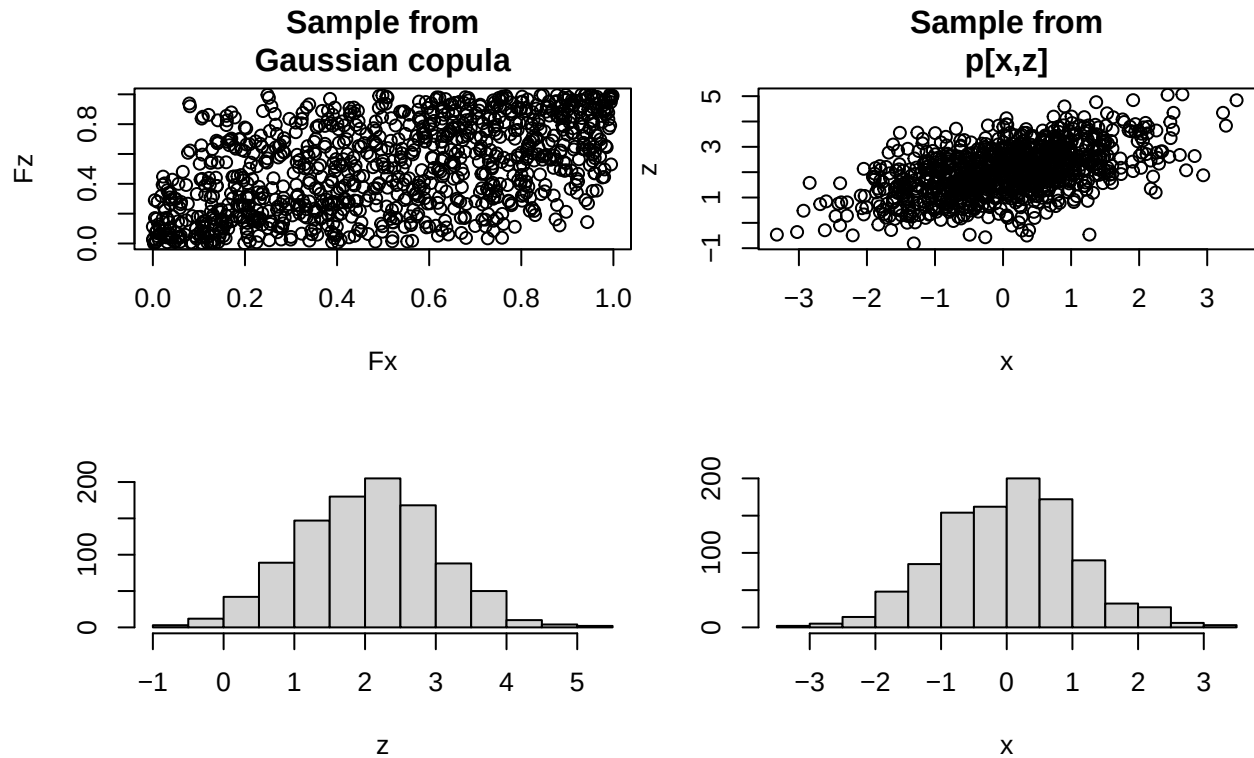


Figure 17: Top left: sample from a Gaussian copula with correlation parameter $\rho = 0.5$. Top right: sample from the joint distribution determined by this copula when the marginals for x and z are $N[0,1]$ and $N[2,1]$. Bottom row: marginal samples.

```
mvN.NG    <- mvdc( cpN, margins = c("norm", "gamma"),
                  paramMargins = list( list( mean =0, sd   =1 ),
                                       list( shape=4, scale=2 ) ) )
```

```
sample.mvN.NG <- rMvdc( n, mvN.NG )
x.N.NG <- sample.mvN.NG[,1] ; z.N.NG <- sample.mvN.NG[,2]
```

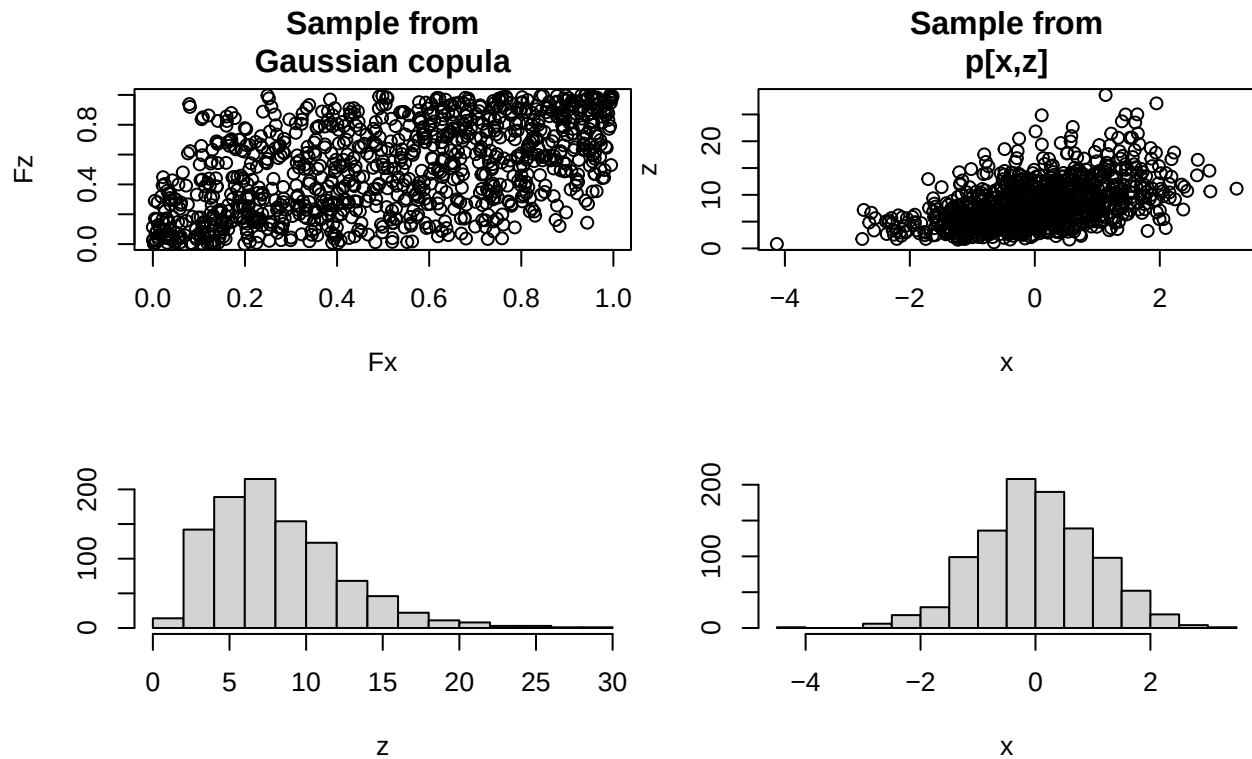


Figure 18: Sampling from a Gaussian copula with correlation parameter $\rho = 0.5$ combined with $N[0,1]$ and $\text{Gamma}[4,2]$ marginals for x and z .

```
> pH      V      R  s_pH  s_V  s_R
> 0.150 0.880 0.140 0.011 0.077 0.016
```

```
> pH      V      R  s_pH  s_V  s_R
> 0.150 3.300 0.500 0.011 0.260 0.055
```

```
cpt <- tCopula( param=0.5, df=1 )
```

```
> pH      V      R  s_pH  s_V  s_R
> 0.150 2.600 0.400 0.011 0.390 0.066
```

Copula selection

```
fitC <- function(x,z){ BiCopSelect( ecdf(x)(x)*n/(n+1), ecdf(z)(z)*n/(n+1),
                                   sel="BIC" ) }
fitC( x.N.NN, z.N.NN )
> Bivariate copula: Gaussian (par = 0.53, tau = 0.35)
fitC( x.N.NG, z.N.NG )
```

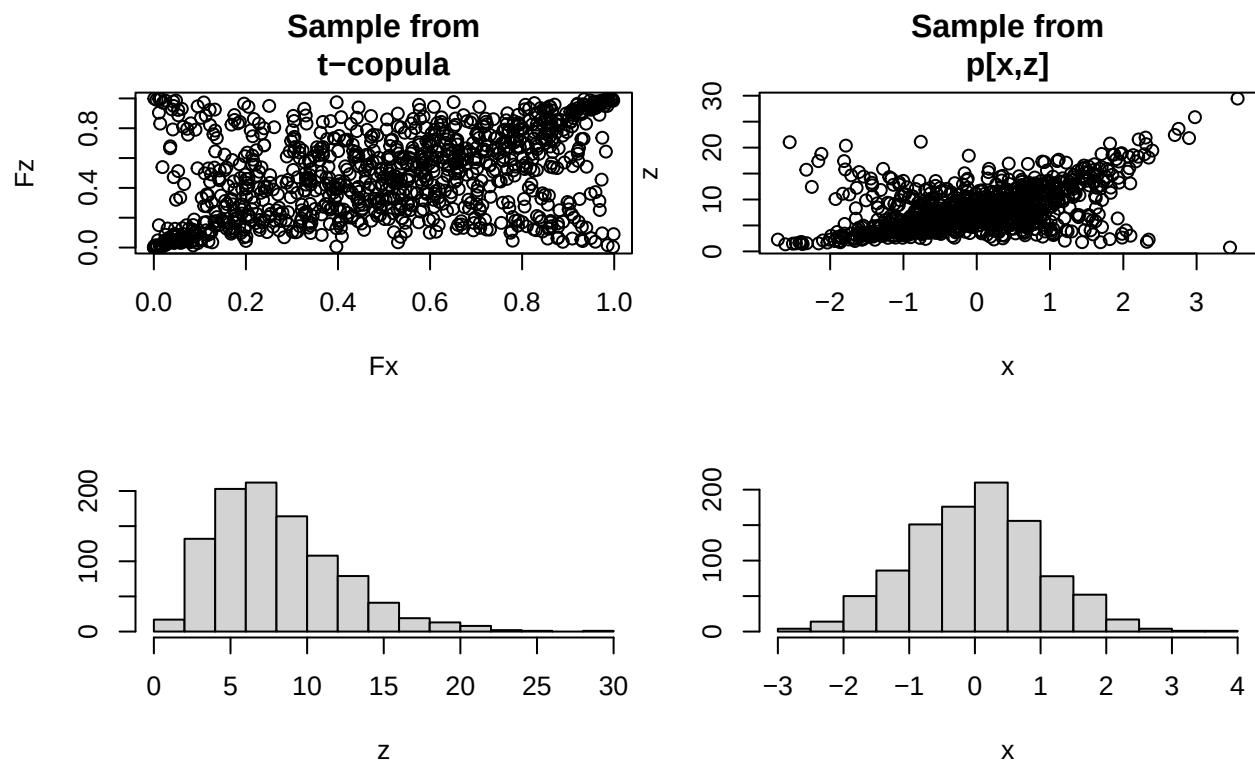


Figure 19: Sampling from a t-copula with correlation parameter $\rho = 0.5$ and $df = 1$ combined with $N[0,1]$ and $\text{Gamma}[4,2]$ marginals for x and z .

```
> Bivariate copula: Gaussian (par = 0.49, tau = 0.33)
fitC( x.t.NG, z.t.NG )
> Bivariate copula: t (par = 0.62, par2 = 2, tau = 0.42)
```

Using copulas in PRA

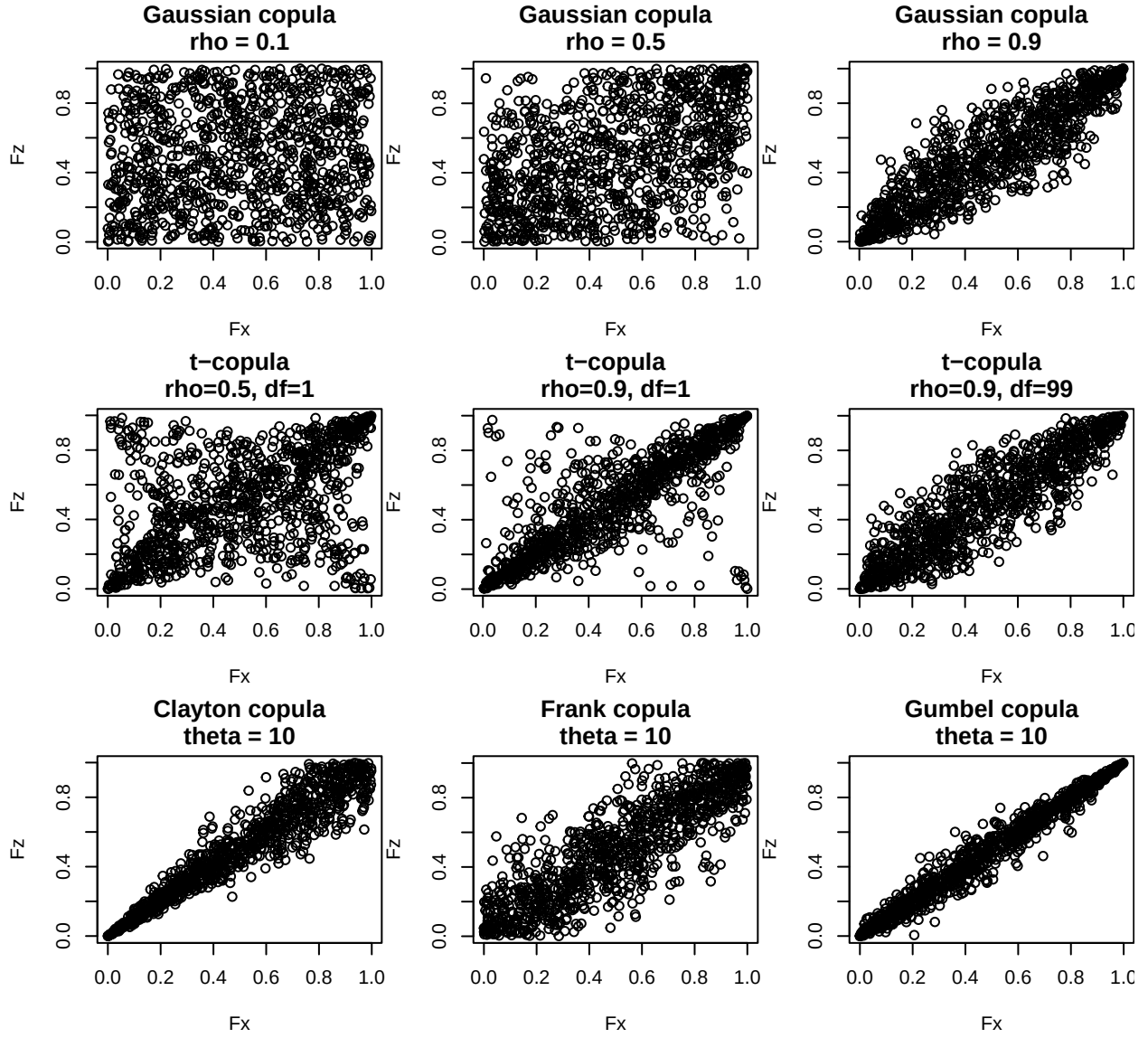


Figure 20: Sampling from different copula families and parameter settings. Top row: Gaussian copulas. Middle row: t-copulas. Bottom row: Clayton, Frank and Gumbel copulas.

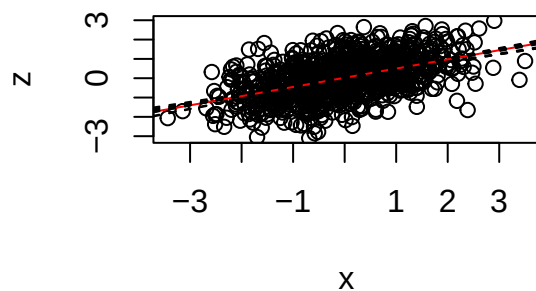
Bayesian model-based PRA

Linear example - Full Bayesian PRA with uncertainty

```
xz <- l_xz.L[[1]] ; x <- xz[,1] ; X <- cbind( 1, x )
z <- xz[,2] ; Vz <- 1 ; Sz <- diag(Vz,n)

# Prior:
mb <- c(0,0) ; Vb <- c(1.e4,1.e4) ; Sb <- diag(Vb)
# Posterior:
Sb_y_LS72 <- solve( solve(Sb) + t(X) %*% solve(Sz) %*% X )
mb_y_LS72 <- Sb_y_LS72 %*% (solve(Sb) %*% mb + t(X) %*% solve(Sz) %*% z)

par(mfrow=c(1,2))
plot( x, z ) ; abline( mb_y_LS72, col="red" )
nsmpl <- 10 ; smpl_b <- rmvnorm( nsmpl, mean=mb_y_LS72, sigma=Sb_y_LS72 )
for( i in 1:nsmpl ) { abline( smpl_b[i,], lty=2 ) }
```



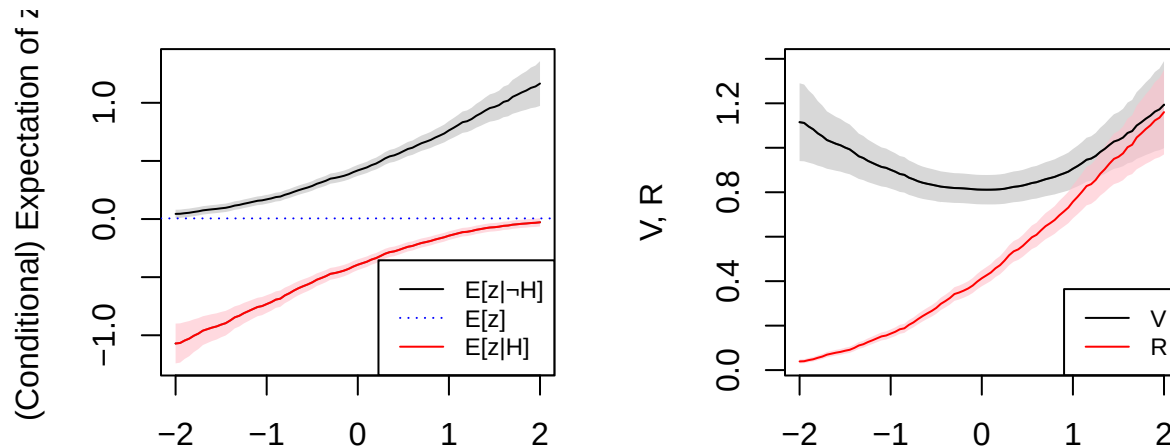
PRA

```
thr <- seq( -2, 2, by=0.05 ) ; n_thr <- length(thr)
n_H <- n_notH <- Ez_H <- Ez_NOTH <- Vz_H <- Vz_notH <- numeric(n_thr)
R <- V <- pH <- s_R <- s_V <- s_pH <- numeric(n_thr)
LCIR <- UCIR <- LCIV <- UCIV <- numeric(n_thr)
LCIzNOTH <- UCIZNOTH <- LCIZH <- UCIZH <- numeric(n_thr)
for(i in 1:n_thr) {
  i_H <- which( x < thr[i] ) ; n_H[i] <- length(i_H)
  i_notH <- which( x >= thr[i] ) ; n_notH[i] <- length(i_notH)
  pH[i] <- n_H[i] / n ; s_pH[i] <- sqrt( pH[i]*(1-pH[i]) / n )
  # V, R from model expectations and UQ from law of total variance
  Ex_H <- mean( x[i_H] ) ; Ez_H[i] <- c(1,Ex_H) %*% mb_y_LS72
  Ex_notH <- mean( x[i_notH] ) ; Ez_notH[i] <- c(1,Ex_notH) %*% mb_y_LS72
  V[i] <- Ez_notH[i] - Ez_H[i]
  R[i] <- pH[i] * V[i]
  Vz_i <- function(i){ t(c(1,x[i])) %*% Sb_y_LS72 %*% c(1,x[i]) + Vz }
  Vz_H[i] <- sum( sapply(i_H, Vz_i) ) / n_H[i] + mb_y_LS72[2]^2 * var(x[i_H])
  Vz_notH[i] <- sum( sapply(i_notH, Vz_i) ) / n_notH[i] + mb_y_LS72[2]^2 * var(x[i_notH])
  s_V[i] <- sqrt( Vz_H[i] / n_H[i] + Vz_notH[i] / n_notH[i] )
  s_R[i] <- sqrt( s_pH[i]^2 * s_V[i]^2 + s_pH[i]^2 * V[i]^2 + pH[i]^2 * s_V[i]^2 )
}
```

```

LCIzNOTH <- Ez_notH - sqrt( Vz_notH / n_notH )
UCIzNOTH <- Ez_notH + sqrt( Vz_notH / n_notH )
LCIzH <- Ez_H - sqrt( Vz_H / n_H )
UCIzH <- Ez_H + sqrt( Vz_H / n_H )
LCIV <- V - s_V ; UCIV <- V + s_V
LCIR <- R - s_R ; UCIR <- R + s_R

```



Nonlinear example - Full Bayesian PRA with uncertainty

```

xz <- l_xz.NL[[1]] ; m <- colMeans(xz) ; S <- cov(xz)
x <- xz[,1] ; z <- xz[,2] ; n <- length(x)

```

```

Model1.Code <- nimbleCode({
  lm.alpha ~ dnorm( 0, sd=100 )
  lm.beta ~ dnorm( 0, sd=100 )
  lm.tau ~ dgamma( 0.01, 0.01 )
  lm.sigma <- 1 / sqrt(lm.tau)
  for(i in 1:ndata){
    lm.mu[i] <- lm.alpha + lm.beta*exp(-x[i])
    z[i] ~ dnorm( lm.mu[i], sd=lm.sigma )
  }
})

```

Checking the MCMC

```

>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
> 0.9795  0.9969  1.0004  1.0004  1.0041  1.0176
>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
> -1.0558 -1.0180 -1.0090 -1.0091 -1.0004 -0.9592
>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
> 0.09547 0.10201 0.10344 0.10353 0.10511 0.11215
>
> Call:
> lm(formula = z ~ exp(-x))
>

```

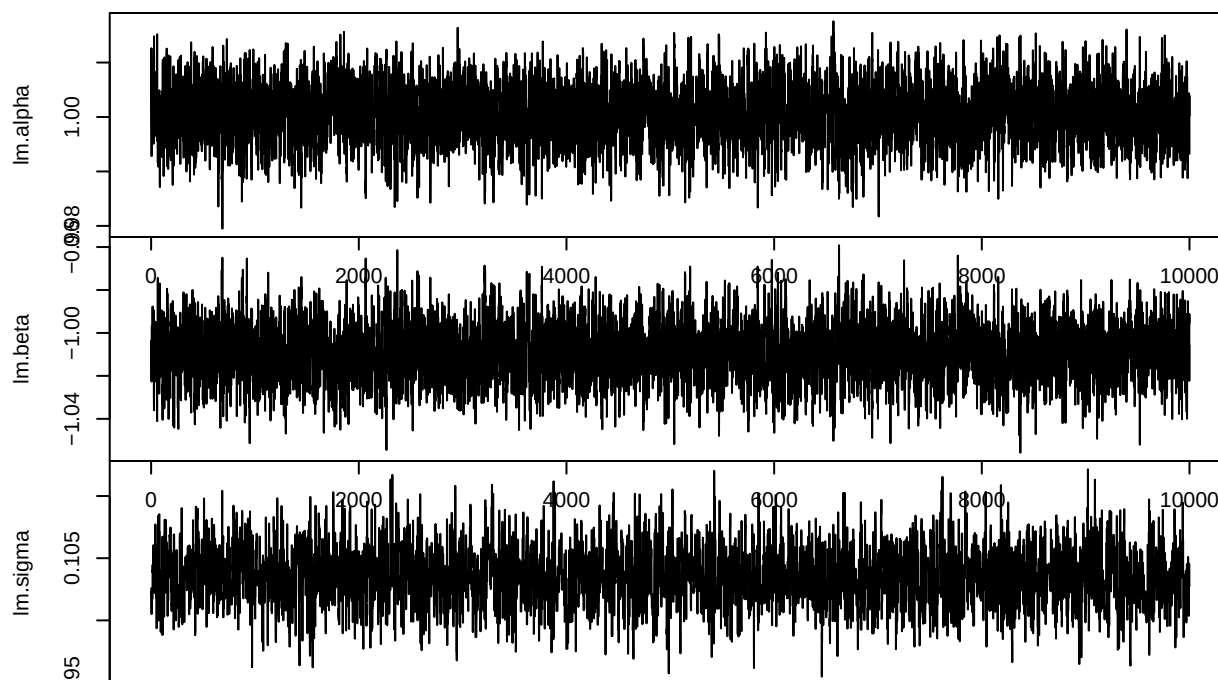


Figure 21: Parameter trace plots generated by Nimble for the nonlinear example.

```
> Residuals:
>      Min       1Q   Median       3Q      Max
> -0.32522 -0.07344 -0.00117  0.07431  0.36544
>
> Coefficients:
>              Estimate Std. Error t value Pr(>|t|)
> (Intercept)  1.000484   0.005226  191.46  <2e-16 ***
> exp(-x)     -1.009282   0.012912  -78.17  <2e-16 ***
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> Residual standard error: 0.1034 on 998 degrees of freedom
> Multiple R-squared:  0.8596, Adjusted R-squared:  0.8595
> F-statistic: 6110 on 1 and 998 DF, p-value: < 2.2e-16
```

PRA

```
n_unc    <- 1e3
itheta   <- sample( 1:ntheta, n_unc, replace=(ntheta<n_unc) )
thr      <- seq( 0.1, 2.9, by=0.05 ) ; n_thr <- length(thr)
EzHj     <- EzNOTHj <- Rj <- Vj <- numeric(n_unc)
EzH      <- EzNOTH <- R <- V <- pH <- numeric(n_thr)
          s_R <- s_V <- s_pH <- numeric(n_thr)
LCIR     <- UCIR <- LCIV <- UCIV <- numeric(n_thr)
LCIzNOTH <- UCIzNOTH <- LCIzH <- UCIzH <- numeric(n_thr)
lm.alpha <- theta[,1] ; lm.beta <- theta[,2] ; lm.sigma <- theta[,3]
for(i in 1:n_thr) {
  i_H <- which( x < thr[i] ) ; n_H <- length(i_H)
```

```

pH[i] <- n_H / n ; s_pH[i] <- sqrt( pH[i]*(1-pH[i]) / n )
for(j in 1:n_unc) {
  zj <- lm.alpha[itheta[j]] + lm.beta[itheta[j]] * exp(-x) +
    rnorm( n, 0, lm.sigma[itheta[j]] )
  EzHj[j] <- mean( zj[i_H] ) ; EzNOTHj[j] <- mean( zj[-i_H] )
  Vj[j] <- EzNOTHj[j] - EzHj[j]
  Rj[j] <- pH[i] * Vj[j] }
R[i] <- mean( Rj ) ; V[i] <- mean( Vj )
s_R[i] <- sd ( Rj ) ; s_V[i] <- sd ( Vj )
EzH[i] <- mean( EzHj ) ; EzNOTH[i] <- mean( EzNOTHj )
qu <- function( z, q=0.025 ){ quantile( z, q, na.rm=T ) }
LCIzNOTH[i] <- qu( EzNOTHj ) ; UCIZNOTH[i] <- qu( EzNOTHj, 0.975 )
LCIzH[i] <- qu( EzHj ) ; UCIZH[i] <- qu( EzHj , 0.975 )
LCIV[i] <- qu( Vj ) ; UCIV[i] <- qu( Vj , 0.975 )
LCIR[i] <- qu( Rj ) ; UCIR[i] <- qu( Rj , 0.975 )
}

```

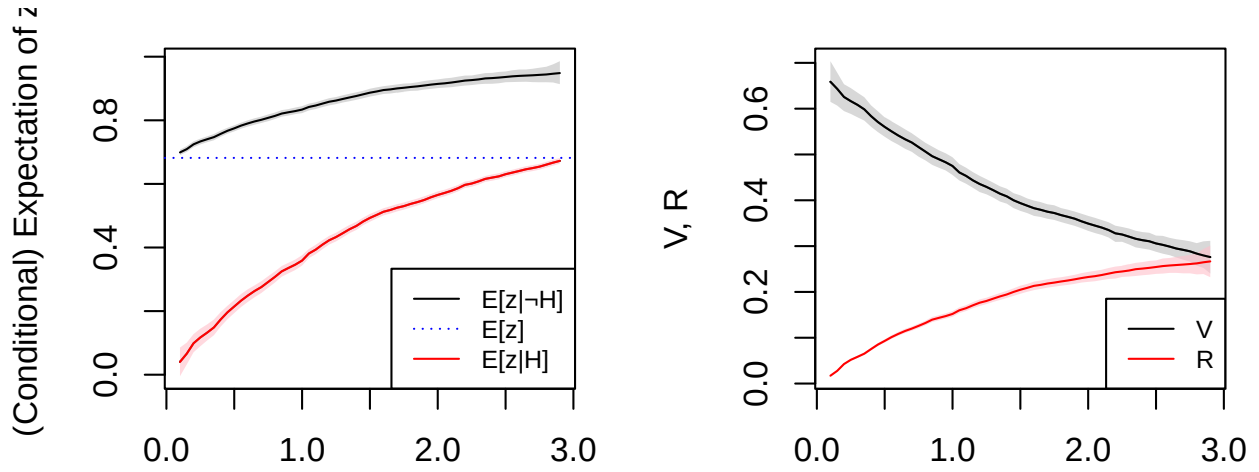


Figure 22: PRA on nonlinearly related x and z , with UQ following Bayesian modelling with Nimble.

Advantages of the Bayesian modelling approach

Sampling-based multi-threshold PRA: Gaussian linear example

```
PRAm <- function( x, z, thr=-1:1 ) {
  n <- length(x) ; n_thr <- length(thr)
  H <- vector("list",n_thr)
  n_H <- pH <- V <- R <- s_pH <- s_V <- s_R <- rep(NA,n_thr)
  H[[1]] <- which( x < thr[1] ) ; n_H[1] <- length(H[[1]])
  for(i in 2:n_thr) { H[[i]] <- which( thr[i-1] <= x & x < thr[i])
    n_H[i] <- length(H[[i]]) } ; n_notH <- n - sum(n_H)
  H.all <- which( x < thr[n_thr] )
  pH <- n_H / n ; s_pH <- sqrt( pH*(1-pH) / n )
  Ez_notH <- mean( z[-H.all] ) ; s_Ez_notH <- sqrt( var(z[-H.all] ) / n_notH )
  for(i in 1:n_thr) {
    Ez_Hi <- mean( z[ H[[i]] ] ) ; s_Ez_Hi <- sqrt( var(z[ H[[i]]]) / n_H[i] )
    V[i] <- Ez_notH - Ez_Hi ; s_V[i] <- sqrt( s_Ez_notH^2 + s_Ez_Hi^2 ) }
  R <- pH * V
  s_R <- sqrt( s_pH^2 * s_V^2 + s_pH^2 * V^2 + pH^2 * s_V^2 )
  R.sum <- sum(R) ; pH.sum <- sum(pH) ; V.wsum <- R.sum / pH.sum
  return( list( sum = c( pH.sum=pH.sum, V.wsum=V.wsum, R.sum=R.sum ),
    seq = cbind( thr, pH, V, R, s_pH, s_V, s_R ) ) )
}
```

```
xz <- l_xz.L[[1]] ; x <- xz[,1] ; z <- xz[,2]
```

```
thr <- -1:-0 ; pram2 <- PRAm(x, z, thr)
```

```
> Overall values: pH.sum = 0.509 ; V.wsum = 0.855556 ; R.sum = 0.435478
> Vector values pH: 0.182 0.327
> Vector values V : 1.148718 0.6923893
> Vector values R : 0.2090667 0.2264113
```

```
thr <- seq(-2,1,0.1) ; pram30 <- PRAm (x, z, thr)
```

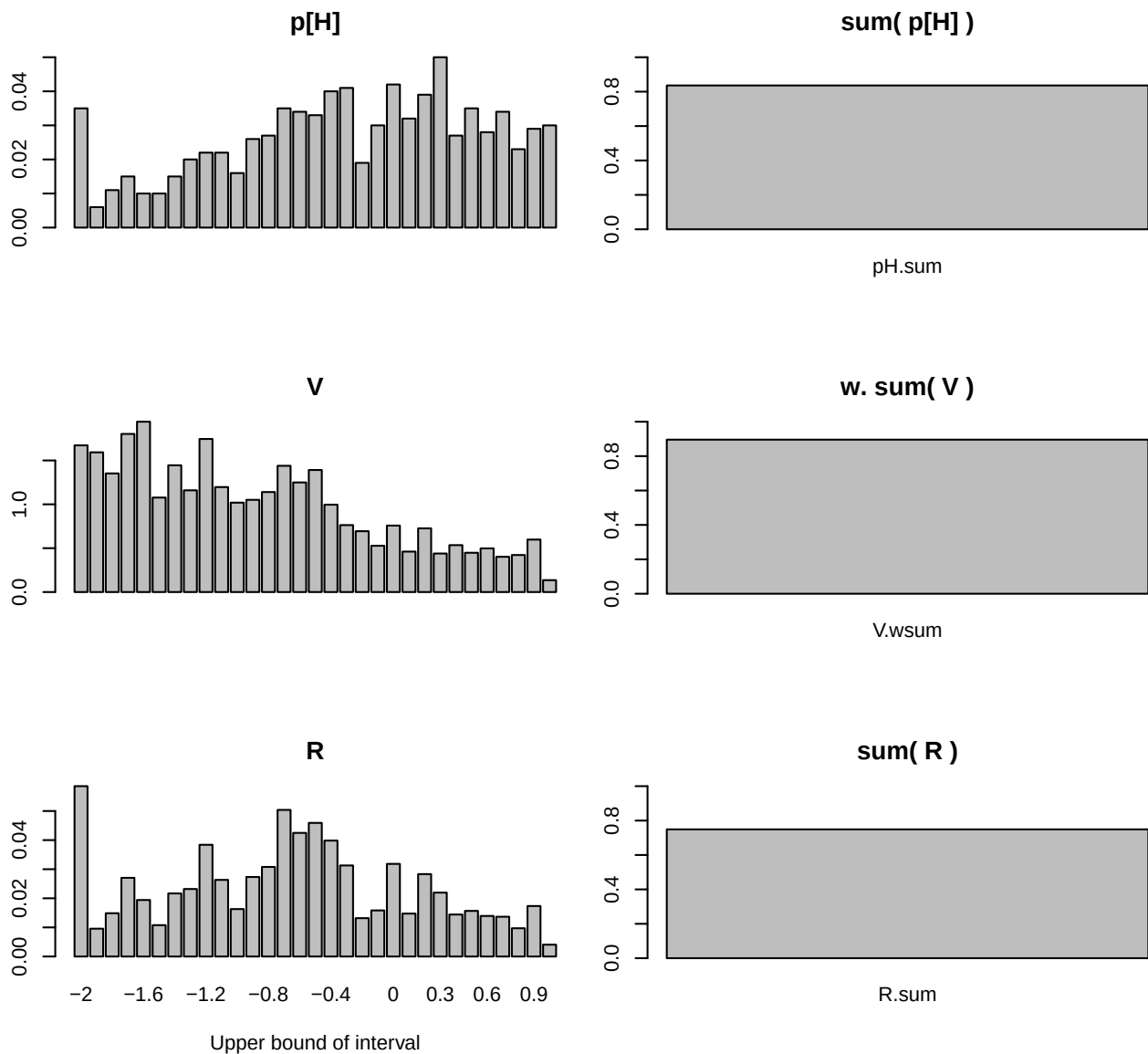


Figure 23: Multi-threshold PRA applied to a Gaussian linear data set. Left: results for 31 hazard-levels. Right: overall (summed) values for all hazard levels combined.

Distribution-based continuous PRA: Gaussian linear example

```
p <- px # Gaussian density function as defined above
mx <- mz <- 0 ; rxz <- 0.5
Ez_x <- function(x){ mz + (x-mx)*rxz }
v <- function(x,thr=0) { Ez_xhi_NI(thr) - Ez_x(x) }
r <- function(x,thr=0) { p(x) * v(x,thr) }

thr <- 1
p.seq <- p(x.seq) ; v.seq <- v(x.seq,thr) ; r.seq <- r(x.seq,thr)
```

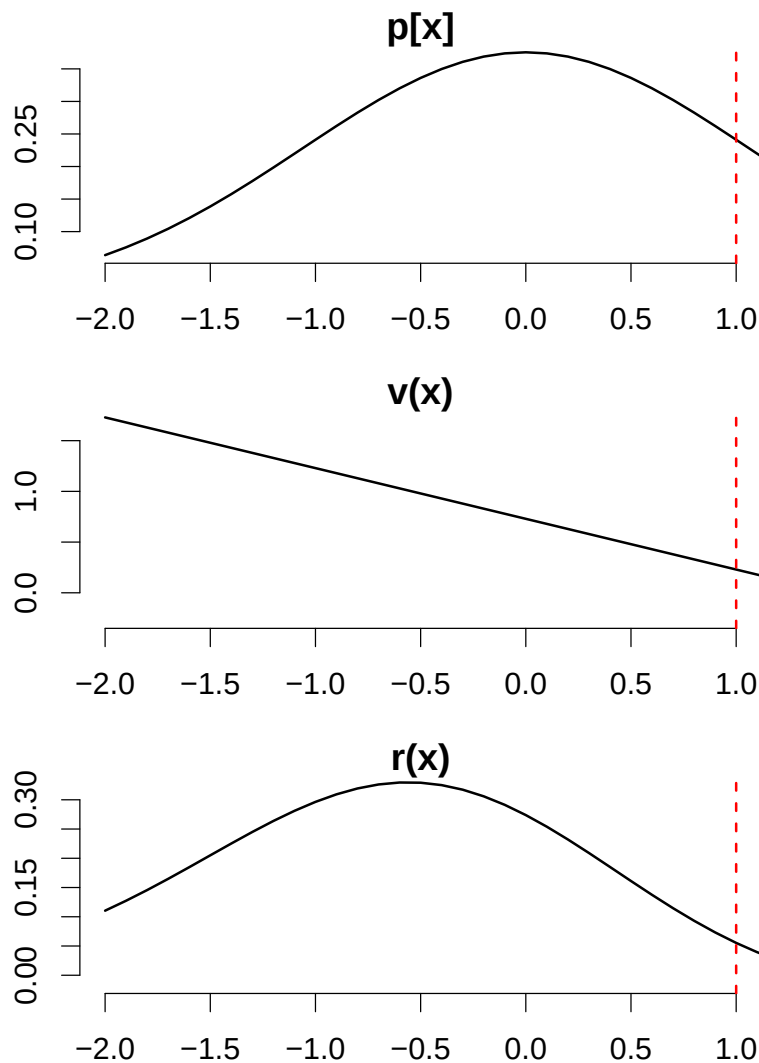


Figure 24: Continuous PRA applied to a bivariate Gaussian dsitribution.

Categorical PRA with other splits than for threshold-levels: spatio-temporal example

Spatio-temporal environmental data: $x(s, t)$

```
GP.AR <- function( s0, s, x, Vx.s, phi, x0past=0, Vx.t=0, alpha=0 ) {
  ds <- as.matrix( dist(s) )
  rx <- exp( -ds/phi )
  ds0 <- sapply(1:length(x),function(i){dist(rbind(s0,s[i,]))})
  r0 <- exp( -ds0/phi )
  m0 <- t(r0) %*% solve(rx) %*% x + alpha * x0past
  V0 <- Vx.s * (1 - t(r0) %*% solve(rx) %*% r0 ) + Vx.t
  return( c( m0=m0, V0=V0 ) ) }
```

```
ns1 <- 8 ; ns2 <- 8 ; nt <- 9
xst <- array( NA, dim=c(ns1,ns2,nt) )

Vx.s <- 0.5 ; phi <- 1
Vx.t <- 0.5 ; alpha <- 0.5
```

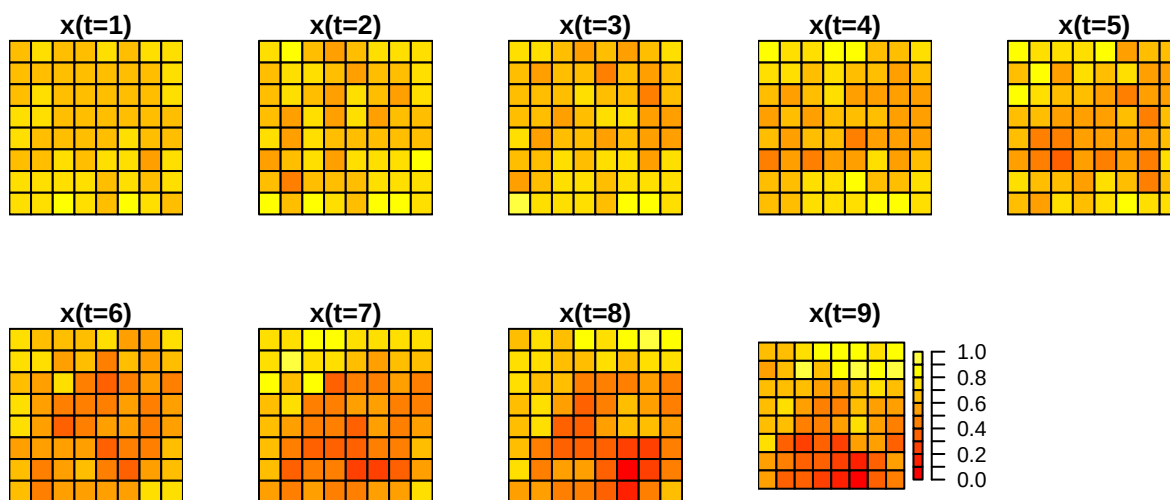


Figure 25: Evolution of the environmental variable $x(s, t)$ over time.

Spatio-temporal system data: $z(s, t)$

```
fz <- function( x, xpast=1, k=10 ) {
  1 / (1 + exp( -k * (x + xpast/2 - 0.5) )) }
```

Single-category single-threshold PRA for the spatio-temporal data

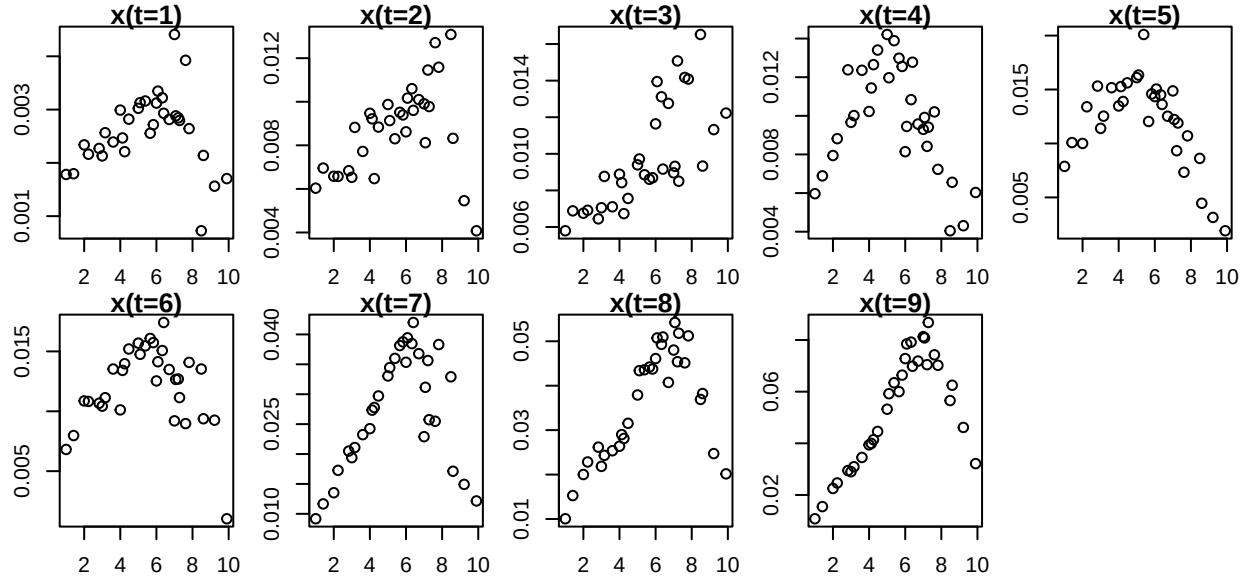


Figure 26: Spatial variograms for $x(z, t)$.

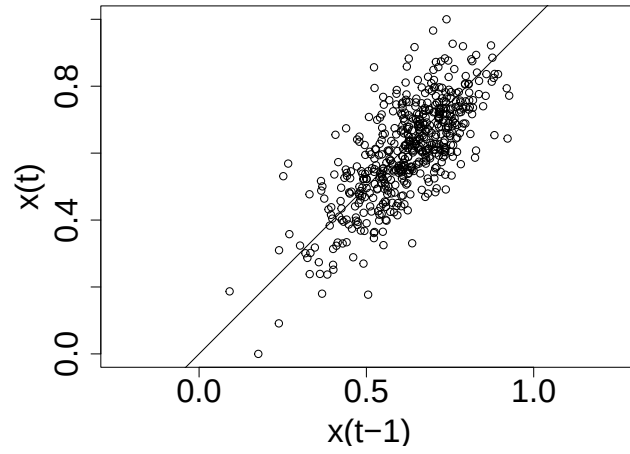


Figure 27: Temporal correlation of the environmental variable.

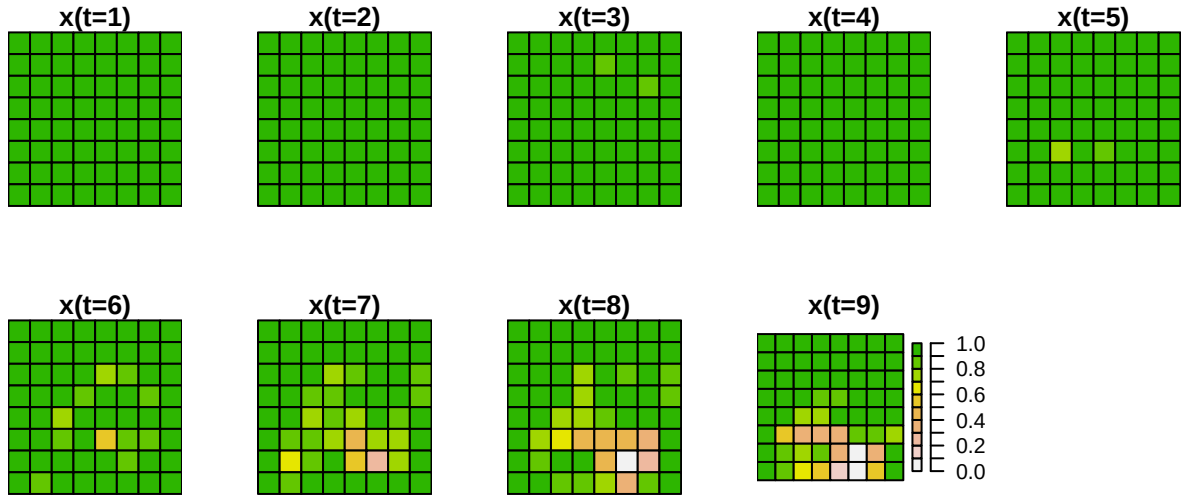


Figure 28: Evolution of the response variable $z(s, t)$ over time.

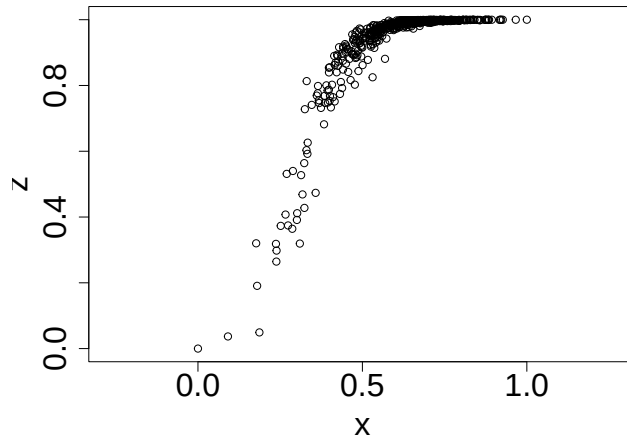


Figure 29: Responses of z to x for all locations and times.

```
thr.xst <- quantile( xst, pnorm(-1) )
```

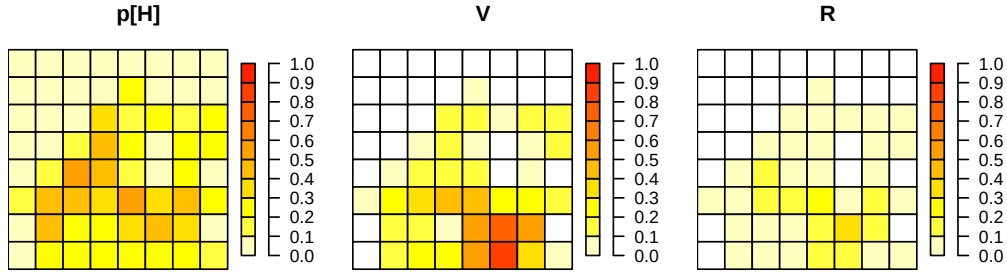


Figure 30: PRAs for all cells in a square region. Each of the 64 PRAs was based on a single cell's time series of (x, z) .

Two-category single-threshold PRA for spatio-temporal data

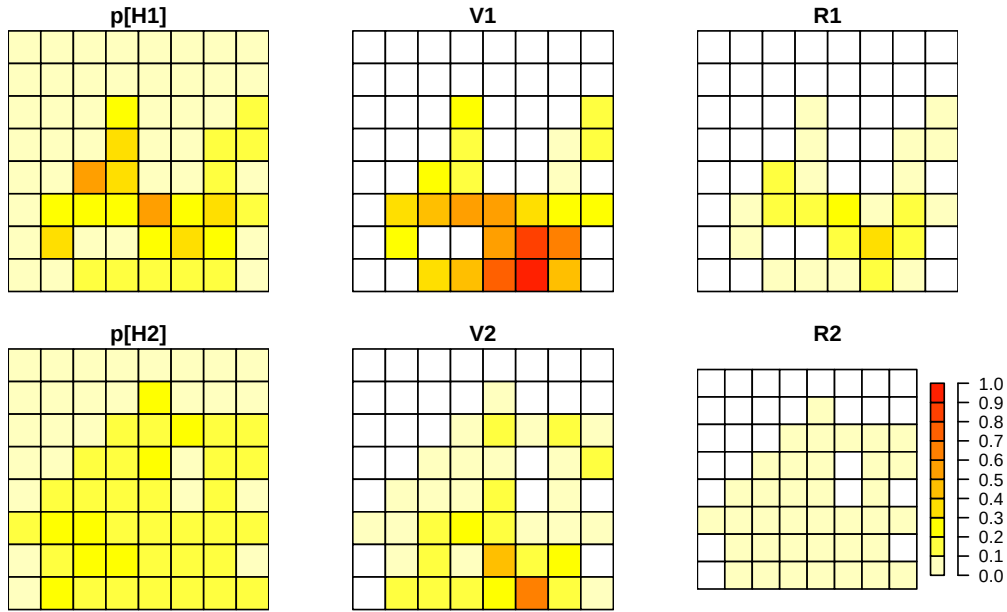


Figure 31: Two-category PRAs for all cells in a square region. Top row: category 1 droughts (preceded by drought in the time step before). Bottom row: category 2 droughts (not preceded by drought).

Three-component PRA

Three-component PRA for spatio-temporal data

```
> Three-component PRA:  
>   Q = 37 ; pH = 0.2762763 ; V = 0.2604741  
>   Rintensive = 0.07196282 ; Rextensive = 2.662624
```

Country-wide application of three-component PRA

UQ for three-component PRA

$$\sigma_{R_e} = \sqrt{\sigma_Q^2 \sigma_{R_i}^2 + \sigma_Q^2 E[R_i]^2 + \sigma_{R_i}^2 E[Q]^2}.(\#eq : sigmaRe) \quad (25)$$